Prototype/ Alpha HVLP-10/20

Applications Reference Manual

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Partner with Performance™

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Table of Contents

Chapter 1: Introduction 1-1

About HVLP-10/20 documentation 1-2

This version of the manual 1-2

Copyright 1-2

Scope of this manual 1-2

Related documents 1-2

Drawings and units 1-2

Warnings, cautions and notes 1-2

Product identification label 1-4

Technical support 1-4

Product warranty 1-4

Chapter 2: About the HVLP-10/20 2-1

Introduction 2-1

Standard features and capabilities 2-2

Intended use of the HVLP-10/20 2-2

Available accessories 2-2

Overview of a motor control application 2-3

Principles of operation 2-4

Functional description 2-4

HVLP-10/20 electrical block diagram 2-4

Interfaces 2-5

General operation 2-5

Safety and protective functions 2-7

General 2-7

On-Highway Vehicles 2-8

HVLP-10/20 Fault detection and handling 2-8

Chapter 3: Installation 3-1

.....3-1

Mounting HVLP-10/20.....3-2

Location.....3-2

Orientation.....3-2

Equipment required:.....3-3

HVLP-10/20 Cooling requirements.....3-3

HVLP-10 – Air Cooling.....3-3

HVLP-20 – Liquid Cooling.....3-4

EMC guidelines.....3-5

General measures.....3-5

Measures required for specific signals.....3-7

Additional measures.....3-8

Problems to avoid.....3-8

Connecting power cables.....3-9

Battery and motor connections.....3-9

Mating and Unmating the HV connectors.....3-10

HVIL Wiring - Preliminary.....3-12

Cable sizes.....3-13

Fuse rating and selection.....3-16

Signal wiring.....3-17

Signal wire sizes.....3-17

CANbus termination.....3-17

Signal connections.....3-18

Control 0V Connections.....3-25

Chapter 4: Specification 4-1

Electrical.....4-1

Input voltage – control supply.....4-2

Input voltage – DC Link supply.....4-2

Motor Output protection 4-2
Output ratings 4-3
CAN interface 4-3
Control inputs and outputs 4-4
Isolation 4-4
Regulatory compliance 4-6
Mechanical 4-6
Operating environment – HVLP-20 (Liquid Cooled) 4-6
Operating environment – HVLP-10 (Air Cooled) 4-7
Shock and vibration 4-7
Weight 4-7
Dimensions HVLP-10/20 4-8

Chapter 5: System design 5-1

Selecting the HVLP-10/20 model 5-2
Circuit configuration 5-2
Auxiliary components 5-3
Main Contactor and Precharge circuit 5-3
Digital Outputs from HVLP-10/20 5-4
Motor speed sensor (AB encoder) 5-5
Motor commutation sensor 5-6
Initial power up sequence 5-10
Checks prior to power up 5-10
Checks after power is applied 5-10
Discharge sequence after power down 5-11
Controller discharge profiles 5-11

Chapter 6: Configuration 6-1

Introduction 6-1
DVT configuration tool 6-2
DVT functionality 6-2

Saving, duplicating and restoring a node's configuration 6-2
Data Logging 6-3
CANopen Configuration 6-3
CANopen protocol 6-3
Object Dictionary 6-3
Network Configuration 6-4
H-Protocol 6-6
J1939 Compatibility 6-7
I-Protocol 6-8
J1939 Compatibility 6-8
Configuration process overview 6-9
Access authorization 6-9
How NMT state affects access to parameters 6-9
Motor characterisation 6-10
Determining Motor Parameters 6-10
Speed Limit Tuning 6-10
I/O configuration 6-10
Encoder 6-11
Digital inputs 6-12
Analog inputs 6-13
Digital (contactor) outputs 6-14
Vehicle performance configuration 6-15
Requirements 6-15
Safety Interlocks 6-15
Driveability Features 6-20
Acceleration and braking 6-22
Footbrake 6-22
Steering inputs – twin driving motor systems 6-22
Driveability profiles 6-24
Smoothing Output Torque 6-27
Scenarios 6-28
Controlled roll-off 6-28

Hill hold 6-29
Inching 6-29
Belly Switch 6-29
Drivability select switches 6-30
Economy 6-30
Seat Regen Braking 6-30
Pump configuration 6-30
Power steer configuration 6-32
Vehicle features and functions 6-32
Line contactor 6-33
Electro-mechanical brake 6-33
Brake Lights 6-34
Horn 6-34
Vehicle Speed Calculation 6-35
Distance Calculation 6-35
Service Indication 6-35
Controller heatsink / motor cooling fan 6-36
Motor over-temperature protection 6-36
Battery protection 6-37
Battery Current Limit 6-38
Displays 6-39
Chapter 7: Monitoring HVLP-10/20 7-1
Reading status variables 7-2
Motor Data 7-2
Customer Input Data 7-2
Device Data 7-2
Identification and version 7-2
Supply monitoring 7-3
Hours counters 7-3
Logging 7-3
FIFO event logs 7-3

Event counters 7-4
Operational monitoring 7-4
CANopen abort code 7-4
Faults and warnings 7-6
Introduction 7-6
Fault identification 7-6
Fault list 7-6
Upgrading the controller software 7-6

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Chapter 1: Introduction

About HVLP-10/20 documentation

This version of the manual

This version of the HVLP-10/20 manual replaces all previous versions. Sevcon has made every effort to ensure this document is complete and accurate at the time of printing. In accordance with our policy of continuing product improvement, all data in this document is subject to change or correction without prior notice.

Copyright

This manual is copyrighted 2015 by Sevcon. All rights are reserved. This manual may not be copied in whole or in part, nor transferred to any other media or language, without the express written permission of Sevcon.

Scope of this manual

The Application Reference Manual provides important information on configuring motor control systems using HVLP-10/20 controllers as well as details on sizing and selecting system components, options and accessories.

The manual also presents important information about the HVLP product range.

Related documents

The following documents are available from Sevcon:

- The Object Dictionary providing important information about CANopen communication with HVLP-10/20.
- Device Configuration Files (DCF) and Electronic Data Sheets (EDS) for each HVLP-10/20 model and revision.

Drawings and units

Orthographic illustrations in this manual are drawn in Third Angle Projection. SI units are used throughout this manual.

Warnings, cautions and notes

Special attention must be paid to the information presented in Warnings, Cautions and Notes when they appear in this manual. Examples of the style and purpose of each are shown below:



A WARNING is an instruction that draws attention to the risk of injury or death and tells you how to avoid the problem.



A CAUTION is an instruction that draws attention to the risk of damage to the product, process or surroundings.



A NOTE indicates important information that helps you make better use of your Sevcon product.

Product identification label

If you have a customized product your unique identifier will appear at the end of the Type number. When discussing technical issues with Sevcon always have your product's Type number, Part number and Serial number available.

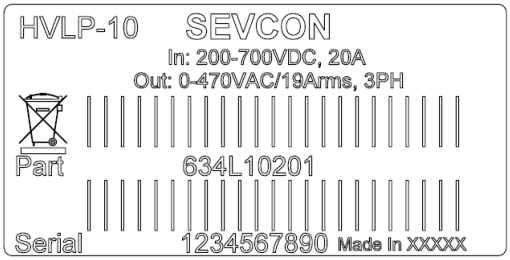


Figure 1 Product Identification label (HVLP-10)

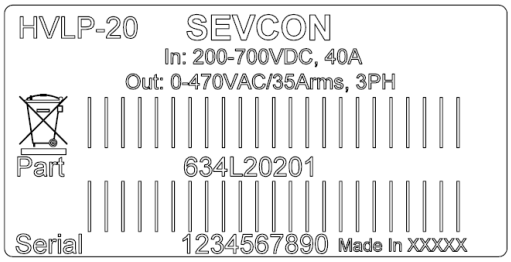


Figure 2 - Product identification label (HVLP-20)

Technical support

For technical queries and application engineering support on this or any other Sevcon product please contact your nearest Sevcon sales office listed on the inside front cover of this manual. Alternatively you can submit enquiries and find the details of the nearest support centre through the Sevcon website, www.sevcon.com.

Product warranty

Please refer to the terms and conditions of sale or contract under which the HVLP-10/20 was purchased for full details of the applicable warranty.

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Chapter 2:
About the HVLP-10/20

Introduction

Sevcon HVLP-10/20 controllers are designed to control 3-phase AC induction motors and Permanent Magnet AC (PMAC) motors in battery and generator powered applications. A range of models are available to suit a wide number of applications and cooling regimes.

The controller has functions to limit the output load current depending on loading conditions, and ambient temperature in which it is operating (temporarily shutting down if necessary). It will also protect itself if the low voltage circuits are incorrectly wired.

Signal wiring and power connections have been designed to be as simple and straight forward as possible. Analogue and digital signal inputs and outputs are provided for switches, sensors, contactors, hydraulic valves and CAN communications. This I/O can also be mapped to the CANbus for control by an external VCU.

Configuration and control of HVLP-10/20 is fully customizable using the latest version of Sevcon's DVT, a Windows based configuration software tool.

The status of the controller can be represented via a signal to a dashboard mounted light for example.

Standard features and capabilities

Intended use of the HVLP-10/20

The HVLP-10/20 motor controller can be used in many applications for auxiliary motor control, such as:

- Agricultural (tractors and implements)
- Light trucks
- Buses
- Marine

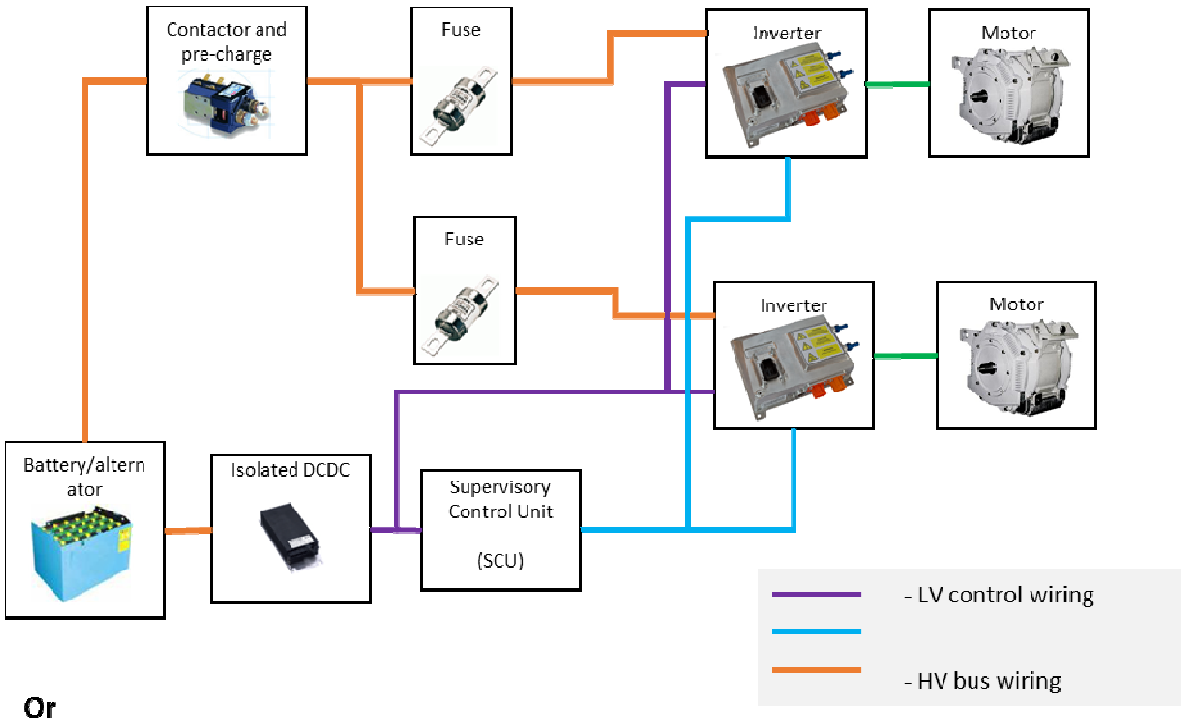
Available accessories

The following accessories are available from Sevcon

- 661/27099 – 48 Way Molex Loose Connector kit of parts (excluding lead wires)
- 184/22147 – 48 Way Molex plus 4M Cable for HVLP used with Resolver sensor
- 184/22148 – 48 Way Molex plus 4M Cable for HVLP used with Sin/Cos sensor
- 184/22149 – 48 Way Molex plus 4M Cable for HVLP used V/F Induction setup
- 184/22145 – 2 Pin TE Plug & 4M Power Cable for HVLP (Battery DC)
- 184/22146 – 3 Pin TE Plug & 4M Power Cable for HVLP (Motor AC)
- DC-DC converters
- DVT - PC based configuration tool

Overview of a motor control application

The main components are shown in Figure 3. In this example there are two auxiliary motors. Communication between the controllers is achieved using the CAN H-protocol and I-protocol.



Or

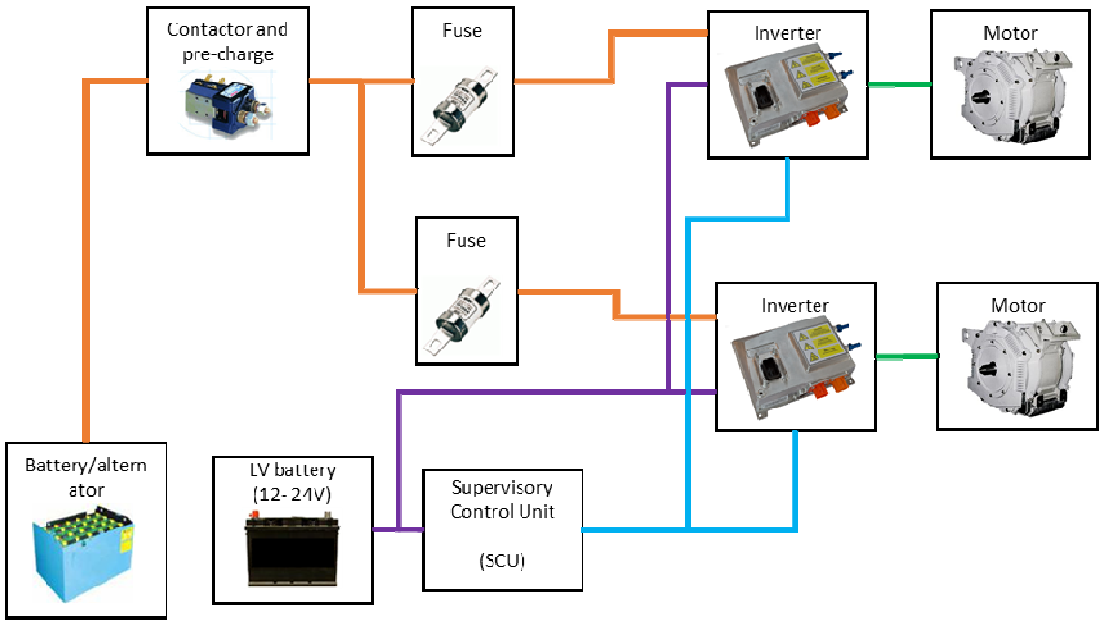


Figure 3 - Vehicle System Components

A fuse, pre-charge circuit and line contactor must be fitted between the battery and the controller power circuits.

Pre-Charge functionality is not built into the HVLP product range. This must therefore be managed external to the HVLP unit.

Principles of operation

Functional description

The main function of HVLP-10/20 is intended to control the power to an ancillary motor (PMAC synchronous or AC Induction).

Measured phase currents and current demands are used as part of a closed-loop control system to calculate the necessary voltage demands for each phase of the motor.

HVLP-10/20 electrical block diagram

The electrical circuit blocks are present in HVLP-10/20 have the following functions:-

- Inverter power stage – this converts the DC battery supply into AC to power the motor. See the following section for more details.
- Gate drives – the low power gate control signal from the control logic is isolated and buffered by these circuits. Output short circuits are also detected by these circuits.
- Current and voltage measurement – these circuits provide an isolated voltage and current measurement for the control logic. Over-current and over-voltage conditions are also detected.
- Control power supplies – these convert a 12V or 24V nominal power supply into the voltage required for the control logic
- Control logic – software runs on microprocessor circuits with input and output circuits for analogue and digital inputs, the encoder, CANbus and analogue outputs.

The high voltage power circuits are fully isolated from the logic control and CAN circuits. All of the required power supplies and control signals for the high voltage system are isolated magnetically or optically from the low voltage system.

All the control inputs and outputs such as the analogue inputs, digital inputs and motor encoder are referenced to the 12/24V logic circuit ground. The CANbus is connected to the control circuit ground.

Do not connect the high voltage circuit to the 12/24V logic circuits directly. If you do this the isolation barrier will be bypassed. This could cause high voltages to be present on circuits which do not have enough protection against accidental touch. Protection of logic circuits against accidental shorts to the high voltage battery relies on this isolation barrier. If a wiring fault was present logic circuits could be damaged.

Power conversion section

The power conversion section of HVLP-10/20 employs a 6-switch IGBT bridge. Excellent electrical and thermal efficiency is achieved by:

- Minimization of thermal resistances.
- Use of the latest IGBT technology
- Internal thermal protection (if temperatures are excessive, switching frequency and output torque may be reduced).
- Overcurrent protection using device characteristics.
- Internal measurement of output current.
- Overvoltage trip in the event of regenerative braking raising battery voltage to unsafe levels.

Interfaces

In addition to its motor control functions, HVLP-10/20 offers many other functions designed to interface with electric systems. A variety of digital and analogue input sources are supported, as listed in 'Signal connections' on page 3-18.

On/off control of contactors is provided by HVLP-10/20. All I/O on the HVLP-10/20 controller is protected against short-circuit to the control logic positive and negative logic supply.

HVLP-10/20 connectivity and interoperability with other system devices (for example A VCU) using a CANbus, the CANopen protocol, Sevcon's H-protocol (providing J1939 compatibility) for communications and I-protocol for I/O configuration is provided. In addition to in-service operation, the CANopen protocol allows the controller to be commissioned using the Sevcon DVT tool.

General operation

The HVLP-10/20 controller functions as a motor control and I/O slave, as shown in Figure 4 - HVLP-10/20 Controller

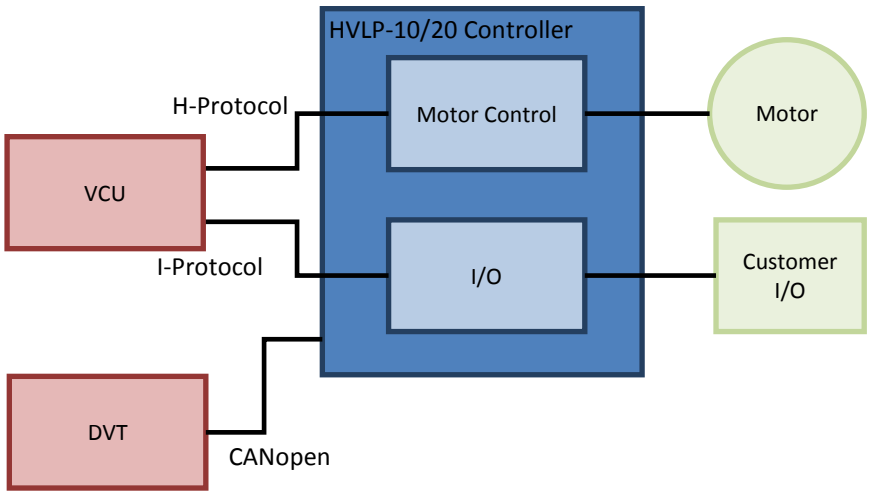


Figure 4 - HVLP-10/20 Controller

The H-protocol allows the VCU to control the state, torque demand, drive/braking torque limits, forward/reverse speed limits and discharge/recharge current limits to be controlled. Refer to Sevcon's H-protocol documentation for more information.

I/O control is managed using the Sevcon I-Protocol. Refer to the I-Protocol documentation for more information.

Safety and protective functions

General

Electric vehicles can be dangerous. All testing, fault-finding and adjustment should be carried out by competent personnel. The vehicle manufacturer's manual should be consulted before any operation is attempted.

The battery or generator must be disconnected before replacing the controller. After the battery has been disconnected, wait for the internal capacitors to discharge to less than 60V before handling the controller or working near exposed terminals.

Hazardous voltages may remain on the controller internally and on exposed power terminals after the main battery power connections and keyswitch power supplies have been removed if the controller is connected to a rotating permanent magnet motor.

Controller discharge profiles on page 5-11 for controller discharge times.

Never connect the controller to a battery with vent caps removed or a generator with an open fuel tank as an arc may occur due to the controller internal capacitance when first connected.

As blow-out magnets are fitted to contactors (except 24V) ensure that no magnetic particles can accumulate in the contact gaps and cause malfunction. Ensure that contactors are wired with the correct polarity to their power terminals as indicated by the + sign on the top moulding.

Do not attempt to open the controller as there are no serviceable components. Opening the controller will invalidate the warranty.

Use cables of the appropriate rating and fuse them according to the applicable national vehicle and electrical codes.

Where appropriate, use of a suitable line contactor should be considered.

Electric vehicles are subject to national and international standards of construction and operation which must be observed. It is the responsibility of the vehicle manufacturer to identify the correct standards and ensure that their vehicle meets these standards. As a

major electrical control component the role of the HVLP-10/20 motor controller should be carefully considered and relevant safety precautions taken. The HVLP-10/20 has several features which can be configured to help the system integrator to meet vehicle safety standards. Sevcon accepts no responsibility for incorrect application of their products.

On-Highway Vehicles

General

This applies to all on-highway vehicles, such as buses and trucks

The installer must ensure an appropriate controller configuration is set to ensure that the vehicle remains in a safe condition, even in the event of a fault.

HVLP-10/20 Fault detection and handling

Fault severity	Controller latched off until	Consequences
Severe (SEV)	Power recycled (turned off then on)	Immediate shut down of the motor control function. Motor power frame is disabled.
Information (INFO)	Not latched	Information faults do not require immediate action, although some de-rating of power or speed may occur.

Table 1 - Fault categories

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Chapter 3: Installation

Mounting HVLP-10/20

Location

The mounting location for the controller should be chosen with care taking into account the following considerations:-

- Do not mount the controller on the outside of a vehicle where it would be accessible to unauthorized personnel.
- Do not mount the controller where it may be susceptible to damage due to minor collisions or impact from road debris.
- Do not mount the controller in the passenger compartment or in the luggage compartment of a road vehicle (See UNECE-REG. 100 clause 5.1.1.3)
- Do not mount the controller in the operator station area of a tractor (see ISO16230-1 2013, clause 5.3
- Although the controller has a high degree of ingress protection, avoid mounting the controller in locations where it may be submerged in water or subjected to long term exposure to jets of water. (Refer to section 4-7 for IP ratings).
- Take note of the thermal and EMC considerations as explained later in this section of the manual.

Orientation

HVLP-20 (Liquid Cooled)

The controller can be mounted in any orientation; however, Sevcon recommend that the controller is located with the connectors pointing downwards as this gives the best environmental protection for the connectors.

It is strongly recommended that the unit baseplate should be mounted to a metal panel which should be substantial enough to provide some additional conductive cooling.

HVLP-10 (Air Cooled)

The controller must be located with the heatsink fins running vertically and preferably in a location with greater than 1m/sec air flow and the ambient temperature does not exceed 45º Celsius (reduced power for temperatures above between 45 and 85º Celsius).

It is strongly recommended that the unit baseplate should be mounted to a metal panel which should be substantial enough to provide some additional conductive cooling.

Mounting hole pattern:

HVLP-10 (Air Cooled)

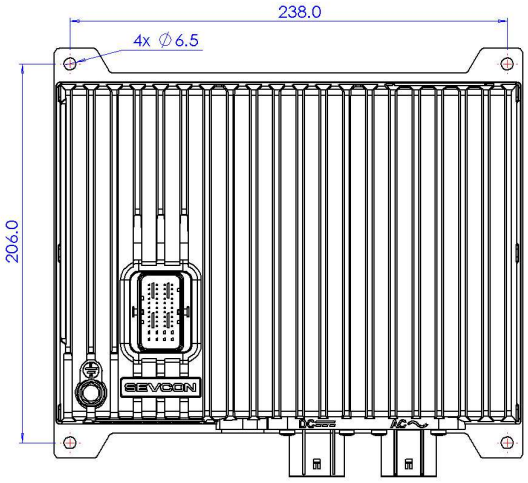


Figure 5 - HVLP-10 Mounting Hole Pattern

HVLP-20 (Liquid Cooled)

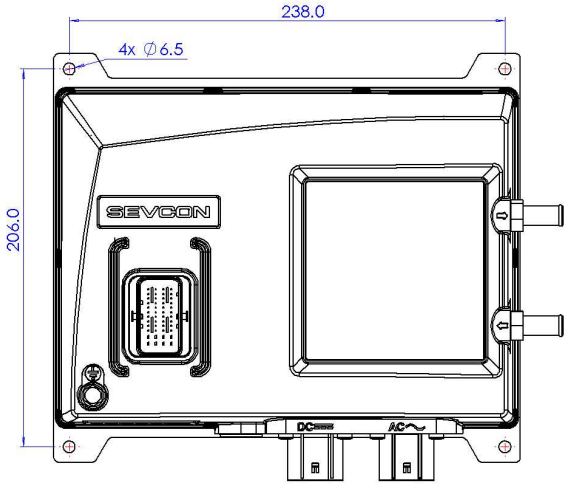


Figure 6 - HVLP-20 Mounting Hole Pattern

(Note: above dimensions expected to be within +/-1.0mm)

The inverter should not be used as a stressed member.

Flatness of mounting surfaces: <0.2mm

⚠ Failure to comply with this flatness specification can cause deformation of the frame and damage to the product.

Equipment required:

- 4 x M6 socket cap head bolts (minimum strength 4.8), nuts and spring washers. Bolts need to be long enough to pass through 5mm of HVLP-10/20 base plate and your mounting surface thickness.
- T hand-socket wrench or Allen key

Recommended torque setting: 10 Nm ± 2 Nm

HVLP-10/20 Cooling requirements

To ensure you get the maximum performance from your HVLP-10/20 controller:

- Keep it away from other heat generating devices on the vehicle
- Maintain its ambient operating temperature below the specified maximum (see 'Operating environment' on page 4-6).

HVLP-10 – Air Cooling

- A minimum of 1m/sec air flow should be available to ensure rated power can be achieved.
- Fins should be kept clean and clear of any dirt and debris.

- Optimum performance will be achieved when mounted in a ventilated location with maximum possible airflow.

HVLP-20 – Liquid Cooling

- A 50/50 water-glycol mixture with a flow-rate of 6 litre/minute should be sufficient to meet the ratings given in this manual.
- Maximum allowable coolant pressure of 1.0bar.
- See Figure 7 showing pressure drop vs flow rate

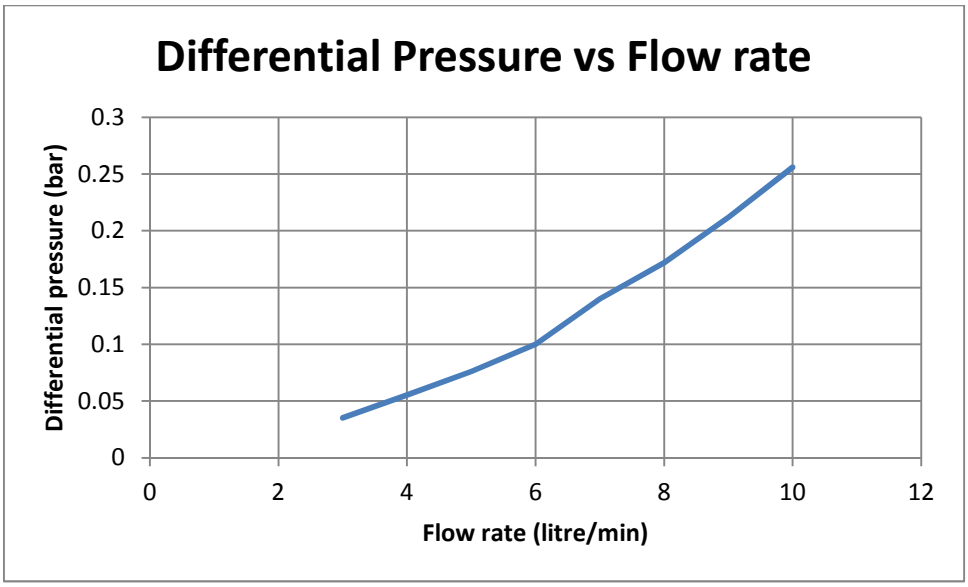


Figure 7: Pressure drop vs. flow rate.

- The HVLP-20 comes fitted with two coolant hose tails. These are installed to the correct torque with a Loctite thread seal and should not be tampered with, changed or removed.
- The hose tails have a straight plain shank (Ø10.5mm) which should accommodate a 10mm (3/8") hose possibly down to 8mm (depending upon the stiffness of the material).
- Arrows on the unit enclosure show the required flow direction.
- Coolant volume is approx. 40ml.
- Hoses should be provided with mechanical fixing with 200mm of the fitting.

EMC guidelines

The following guidelines are intended to help vehicle manufacturers to meet the requirements for Electromagnetic Compatibility. Any high speed switch is capable of generating harmonics at frequencies that are many multiples of its basic operating frequency. It is the objective of a good installation to minimise, contain or absorb the resultant emissions. All wiring is capable of acting as a receiving or transmitting antenna. Arrange wiring to take maximum advantage of the structural metal work inherent in most vehicles. Link vehicle metalwork with conductive braids.

General measures

Power cables

Use shielded power cable for all connections to the motor and the battery. Where cables pass through metal enclosures, such as at the motor and battery; use metal cable glands to connect the cable shield to the enclosure, ensuring that the various enclosures are linked with conductive braid to the vehicle chassis.

Route all cable within the vehicle framework and keep as low in the structure as is practical - a cable run within a main chassis member is better screened from the environment than one routed through or adjacent to an overhead guard. Keep cables short to minimize emitting and receiving surfaces. Shielding by the structure may not always be sufficient - cables run through metal shrouds may be required to contain emissions.

Parallel runs of cables in common circuits can serve to cancel emissions - the battery positive and negative cables following similar paths is an example. Tie all cables into a fixed layout and do not deviate from the approved layout in production vehicles. A re-routed battery cable could negate any approvals obtained.

The cable manufacturers' recommendations for minimum bend radius should always be followed.

Good practice when routing cables is to keep power cables at least 300mm from the signal cables.

Chassis protective earth connection

It is essential that the chassis earth connection on the enclosure is connected to the vehicle or machine chassis using a short cable.

The controller is supplied with an M6 hex-head screw for this purpose.

The protective chassis wire must be 4mm^2 minimum cable cross-sectional area, terminated at the controller end with a suitable M6 ring terminal.

The screw is supplied with a flat washer and spring washer which should all be refitted, and the screw tightened to 5Nm +/- 10%. See diagram below

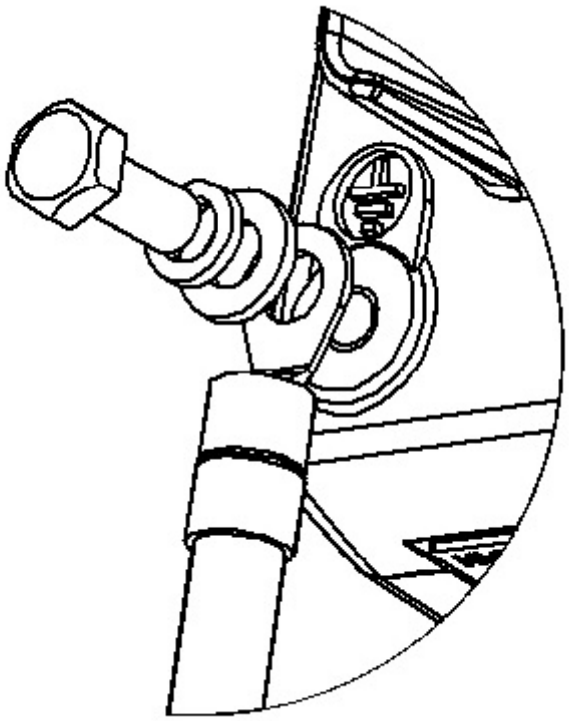


Figure 8 - Protective Earth Connection Assembly

WARNING: Failure to provide a correctly sized protective earth connection could result in an electric shock risk and non-compliance with international electrical safety regulations

The protective earth connection should have a maximum resistance of 0.1Ω when measured at 20A

Notes:

1. The high voltage cable shields should be regarded as EMC shields and are not a substitute for the protective earth connection described above
2. **The vehicle or machine is required to have a ground fault monitor (protection by automatic disconnection to compley with the requirements of section 6.3.3 for IT systems, IEC 60204-1: edition 5.1 2009.)**

Signal cables

Keep all wiring harnesses short and route wiring close to vehicle metalwork. Keep all signal wires clear of power cables and consider the use of shielded cable. Keep control wiring clear of power cables when carrying analogue information - for example, accelerator wiring and speed feedback. Tie all wiring securely and ensure it always follows the same layout.

Measures required for specific signals

High voltage battery/generator power cables

Use shielded cables. Cables should be as short as possible. Minimise the loop area of the B+ and B- wiring. A common mode ferrite choke may be required in some vehicle installations.

Motor power cables

Use shielded cables. Use metal cable glands and connect the shield of the cable to the gland. Cables should be as short as possible. Minimise the loop area of the M1 M2 M3 wiring. A common mode ferrite choke may be required in some vehicle installations.

Motor encoder connection, types AB, UVW and Sin-Cos

The encoder cable must be as short as possible. A multi-core shielded cable should be used. The inner cores should be used for the encoder supply and encoder ground and the AB or UVW or Sin-Cos signals. The cable shield should be connected to the encoder ground terminal. Ensure that the shield does not connect the motor chassis back to the encoder ground terminal to avoid motor power cable current returning through the encoder cable shield.

AB and UVW are differential signal inputs and therefore performance benefits may be seen with twisted pair shielded cables.

Motor encoder connection, resolver types

A resolver is a device consisting of a primary excitation winding (driven by a 10 kHz sinusoidal carrier signal generated by the HVLP-10/20) and two secondary windings where the carrier signal is modulated at the rotational frequency by the sine and cosine of the rotor angle.

The encoder cable must be as short as possible. Separate twisted-pair shielded cables should be used for the primary excitation and the sine and cosine secondary signals. The inner cores should be used for the sinusoidal signals. The cable shield should be connected to the encoder ground terminal. Ensure that the shields do not connect the motor chassis back to the encoder ground terminal to avoid motor power cable current returning through the encoder cable shield.

CANbus

A multi-core shielded cable should be used. The inner cores must be twisted pairs. The twisted pair should be used for CANH and CANL. Ensure that there is a common ground connection for all nodes on the CAN bus. If there is a node on the bus which is galvanically isolated from the HVLP-10/20 controller then the CAN ground on this node must be connected to the HVLP-10/20 controller control OV. The cable shield should be connected to the control OV terminal at the HVLP-10/20 controller

Keyswitch (KL15 – KL30)

HVLP-10/20 offers 2 options for key switch control.

1. Combined ignition / battery control
2. Independent ignition / battery control

Combined ignition / battery control

Connect KL30 and KL15 together via a suitably rated key switch (10A DC) and fuse (7.5A recommended, 10A maximum) to the supply.

When key switch is opened, the controller switches off immediately.

Independent ignition / battery control

Connect KL30 via a fuse (5A recommended, 10A maximum) to the supply. Connect KL15 via a fuse (5A recommended, 10A maximum) and key switch to the supply. KL30 is active at all times. KL15 is the ignition switch.

The two digital outputs, which are rated at 2A each, are both powered from the KL15 supply. For this reason the key switch should be rated at 5A.

Power up /down functionality

At power up, closing KL15 activates the controller as normal. The internal power supplies activate and the controller executes its start-up process and begins communicating on the CANbus.

When KL15 is opened, torque is removed from the motor and all user outputs turn off. When the motor speed has dropped below a configured level the controller prepares to shut-down. After 10s or reception of a SHUTDOWN command on the H-Protocol the internal power supplies are deactivated and the controller shuts down. Closing KL15 again before the shutdown will abort the shutdown process and normal operation will resume.

Both KL30 and KL15 inputs are 12V or 24V nominal. This wiring should be as short as possible. Minimise the loop area of the loop formed by the supply, OV return and any switches. Do not connect additional loads to these wires.

Additional measures

Where it has not been possible to meet the required EMC specifications using the standard measures listed above it may be necessary to use one or more of the following measures:-

- Use of shielded cable for all control connections
- Install additional EMC shielding around the controller and/or motor
- Addition of an LC filter on the keyswitch supply
- A common mode ferrite choke for all the small signal connections will attenuate common mode emissions

Problems to avoid



EMC is a complex subject and on a typical vehicle there are many potential radiators and recievers. Measures taken on a vehicle to improve EMC can unitentionally make the situation worse.

- Beware of devices that are connected to the small signal wiring which have a significant (>10nF) capacitance to vehicle chassis. The capacitance to vehicle chassis can cause currents to flow out of the HVLP-10/20 controller along the signal wiring to the device and back to the controller via the vehicle chassis.
- When using shielded cable beware of generating ground loops in which currents may be induced or which may cause noise currents to flow via unintended paths.

Connecting power cables

See also the section on EMC guidelines.

Battery and motor connections

Note: Early prototypes with Serial numbers ranging from 1405000001 to 1409000008 were manufactured with a different pin allocation on the motor header connector and caution should be exercised if replacing one of these controllers. Before installation, consult with Sevcon if your unit serial number falls within this serial number range.

Cables carrying high AC currents are subject to alternating forces and may require support in the cable harness to avoid long-term fatigue.

- Use shielded power cables sized to suit the controller and application (see Table 3 below)
- TE HVA-280 high voltage connectors
- Cables are supplied pre-assembled to the connectors
- The DC Input cable includes a separate cable (shielded twisted pair) for the HVIL, and the two conductors for the high voltage supply, with an overall shield
- The motor output cable is also shielded, with 3 conductors.
- Ensure that minimum bend radius specifications are complied with for high voltage cables (typ. 5 x cable diameter)
- Ensure cables are clipped within 200mm of the connector location
- Try to prevent cables being flexed relative to the connector outlet as the connector seal can be damaged by this.

All external cables must have orange outer covering to comply with ECE Reg-100 or ISO 16230

Connector	Connector Location No	PCB Location	Wire
DC Link (184/22145) *(Figure 9 - DC-Link Connector Pin Numbering) Key Type B*	1	SK6 (+ve)	Red, 4mm²
	2	SK7 (-ve)	Blue, 4mm²
	3	High Voltage Interlock – not currently in use.	Black, 0.5mm²
	4		
Motor (184/22146) *(Figure 10 - Motor Connector Pin Numbering)*	1	SK8 (M1)	Red (1), 4mm²
	2	SK9 (M2)	Red (2), 4mm²
	3	SK10 (M3)	Red (3), 4mm²

Key Type A			

Table 2 High Voltage Connectors

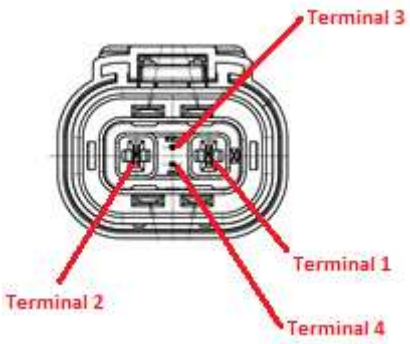


Figure 9 - DC-Link Connector Pin Numbering

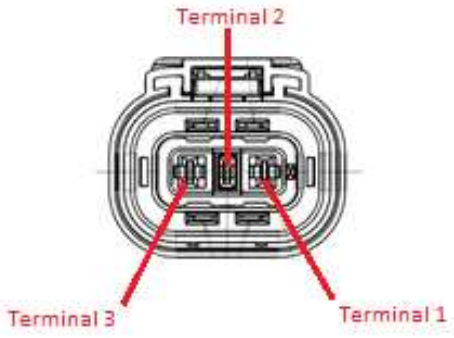


Figure 10 - Motor Connector Pin Numbering

Consider cable routing before making connections.

- Keep cable runs short
- Minimize current loops by keeping positive and negative cables as close together as possible.

Mating and Unmating the HV connectors

Conditions for Mating and Unmating the HV connectors

Connectors should be mated and unmated with the HV power off and in a safe condition, for example by opening a service plug and waiting for 5 minutes to allow for capacitor discharge.

⚠ Under no circumstances should the DC power connector be mated with the DC power supply live, as this would circumvent the pre-charge process and result in a very large inrush current

⚠ **If it is essential to mate or unmate either of the connectors with power applied, this must be done in a clean, dry environment. Attempting to mate or unmate the connectors in a wet environment is extremely dangerous**

Connector Mating

The text below is provided by TE Connectivity (see application specification 114-13305 revE)

- Align plug and header connectors as shown in Figure 11(A)
- Push the connector halves together fully until they stop (two clicks will be heard).
- Verify that the connectors are fully locked with a light tug and by visually verifying the thumb actuation pad has returned to its' upward position as shown in Figure 11(B).

- If present, push the red Connector Position Assurance (CPA) forward from the position shown in Figure 21B until it stops as shown in Figure 11(C).

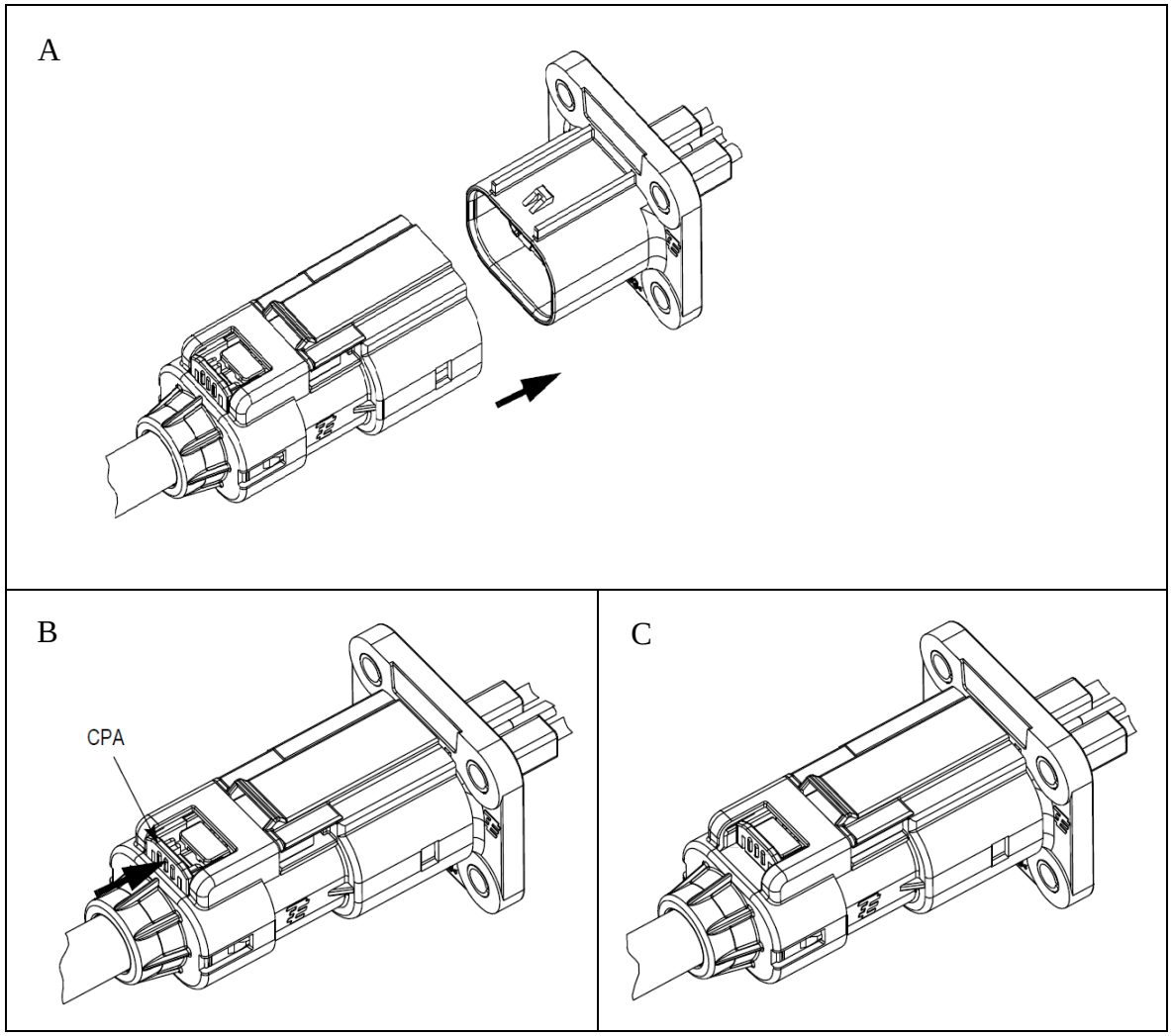


Figure 11 - Connector Mating

Connector Unmating

1. If present, pull the red CPA back from the position as shown in Figure 12 (A) until it stops as shown in Figure 12 (B)
2. Fully depress the thumb actuation pad on the plug connector flexible latch (orange) shown in Figure 12 (B) and pull the plug connector back (~6.2 mm) into the intermediate position. See Figure 12 (C). At this point, in the case of the 2-terminal connector, the HVIL circuit has been opened, but the HV terminals are still making contact.

⚠ Do NOT pull on the wires.

3. Fully depress the actuation pad on the floating latch (black). See Figure 12 (D)..
4. An alternate construction may not require a flat-bladed screwdriver. See Figure 12 (E).
5. Pull the plug connector until it is fully separated from the header. See Figure 12 (F)

⚠

Do NOT pull on the wires.

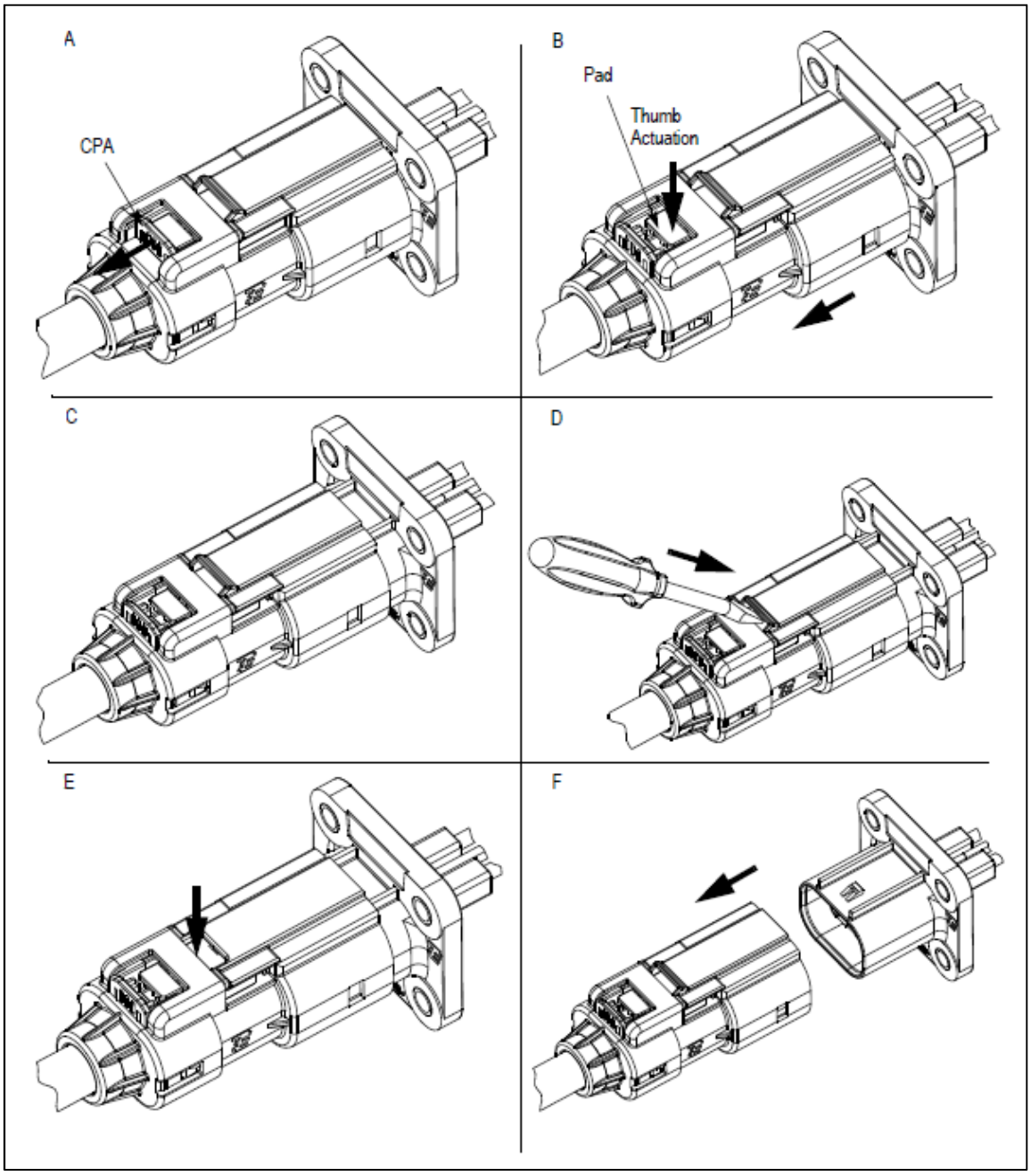


Figure 12 - Connector Unmating

HVIL Wiring - Preliminary

The HVIL (High Voltage Interlock Loop) is included to prevent high voltages from being present on the vehicle or machine in the event of a fault, or if a connection is unplugged.

Devices in the loop are expected to be connected in series, and if the loop opens the high voltage shall be switched off.

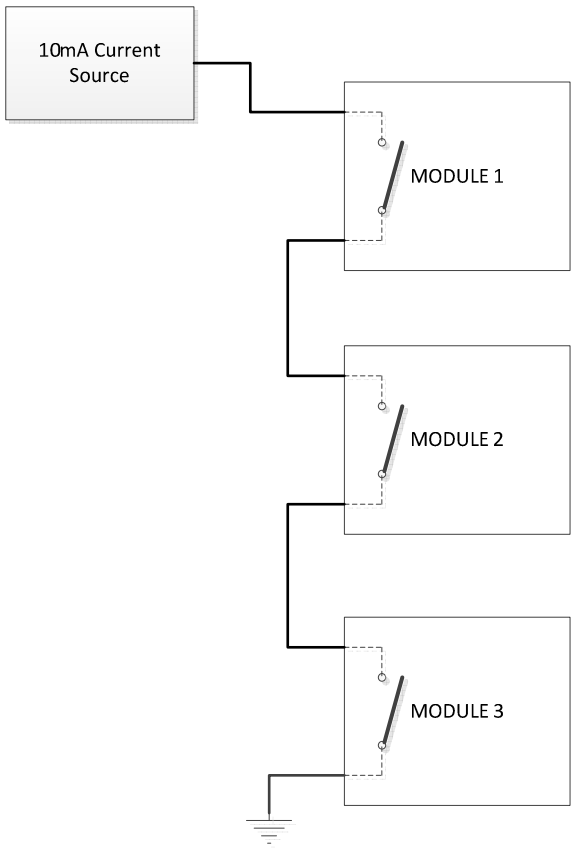


Figure 13 - Representative HVIL configuration.

The HVLP-10/20 controller provides the following HVIL functions:

- HVIL Excitation Supply (10mA current source). This can be used to originate the HVIL if required. Its use is optional
- HVIL Logic Inputs. Two short-circuit protect inputs are provided on the I/O connector which can be connected either way round.
- HVIL passive loop on high voltage DC input connector. If connected a part of the HVIL, the loop will be opened if the DC input connector in unplugged.

HVIL Logic Inputs

The HVIL logic inputs monitor the loop current and indicate to the microcontroller whether the loop is correctly connected.

If the HVIL functionality is enabled it will provide the following functions:

- If the controller detects that the loop has been opened it will cause the controller to shutdown

Cable sizes

When deciding on power cable diameter, consideration must be given to cable length, grouping of cables, the maximum allowable temperature rise and the temperature rating of the chosen cable.

The following table (Table 3) gives guidance on the cable size needed for various currents in shielded power cable, not grouped with other cables, in 30°C ambient with 60°C temperature rise on the cable surface.

HVLP-10/20 average (rms) current	Cable sizes	
	metric	US (approx equivalent)
20 A	2.5 mm^2	12AWG
35 A	4 mm^2	10AWG

Table 3: Guidance on rating of shielded cable

Note that the standard cable (TE supplied with the 3 terminal HVA280) connector conductor area is 2.5mm^2 as shown in the table above. Depending on the motor current the cable conductor temperature may be excessive. See Figure 14

An alternative connector cable combination is available from Sevcon (Order code 184/22146) This uses a 4mm^2 conductor area, and will achieve lower conductor temperature. See Figure 15. This cable/connector solution has not yet been validated by TE and both Sevcon, Huber and Suhner and TE are working to achieve formal validation. Until then, customers must replace the cable/connector whenever the controller is removed/replaced.

An additional factor is the current rating of the connector. At rated maximum operating temperatures (45°C ambient, 60°C coolant) the maximum continuous current that can be accommodated by the connector terminal is 30A using TE supplied cable, and 35A using the Sevcon supplied 4mm^2 cable. (Testing carried out in accordance with EN60512-5-2:2002)

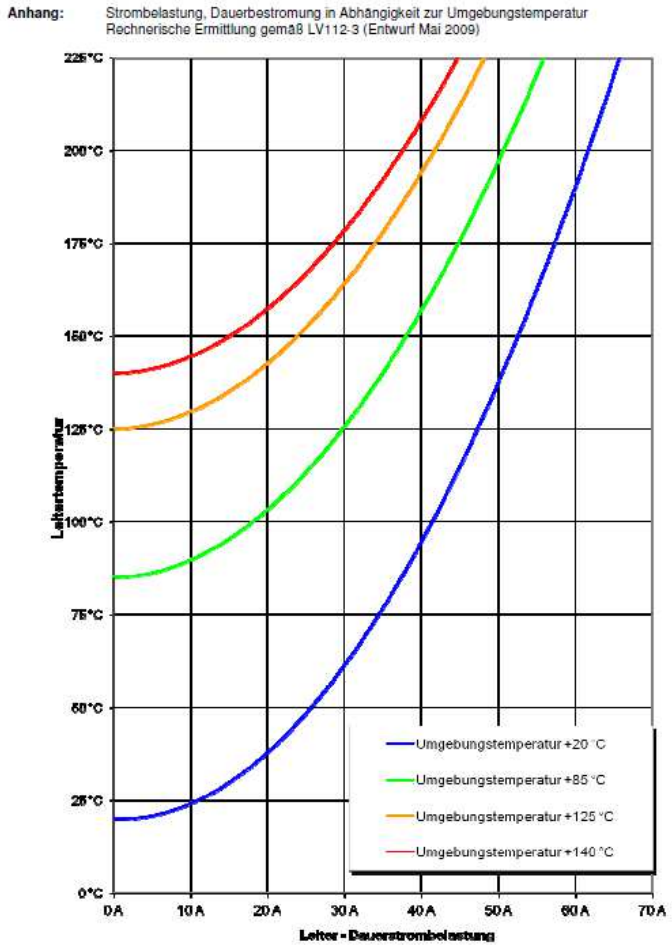


Figure 14 - Coroplast 2.5mm² Cable Conductor Temperature vs. Current

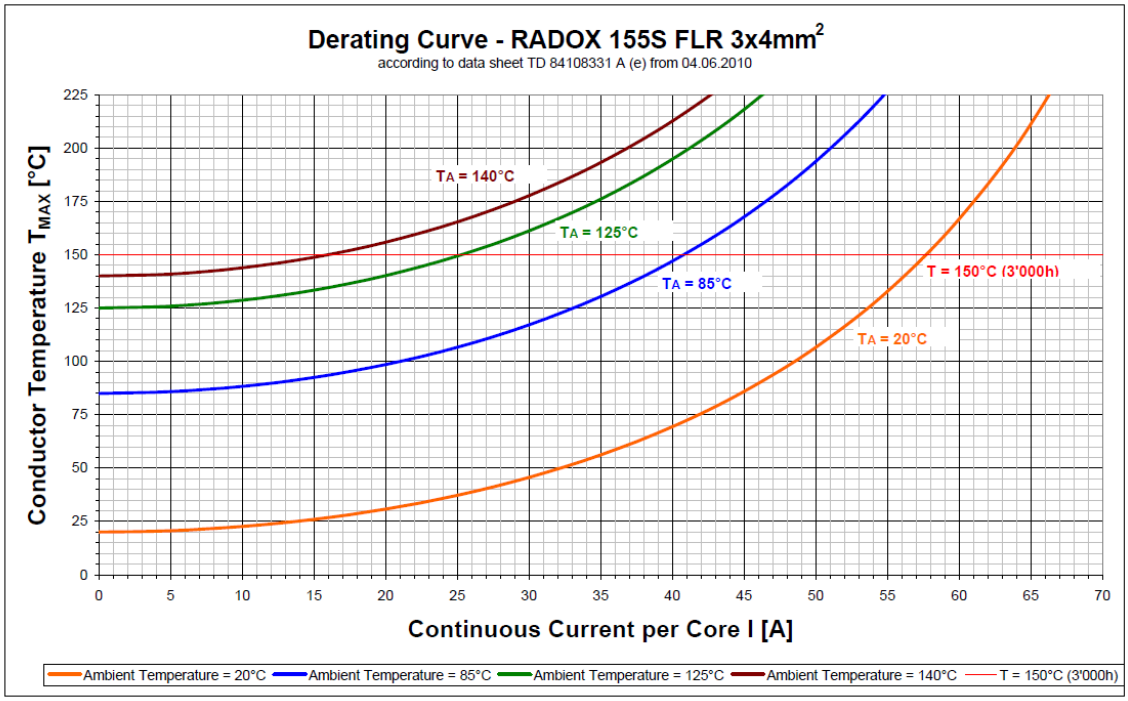


Figure 15 - - H&S Radox 155S 4mm2 Cable Conductor Temperature vs. Current

Fuse rating and selection

See Auxiliary Components (page 5-3)

Signal wiring

Assemble your wiring harness using wire of the sizes recommended below and the Sevcon loose connector kit (p/n 661/27099). The use of twisted pair and in some cases twisted-shielded cables is recommended for the speed sensor and CANbus wiring.

To make a connection, gently push the connector housing onto the appropriate mating half on the HVLP-10/20. Never force a connector. Connectors are keyed to prevent incorrect insertion.

See also the section on EMC.

Signal wire sizes

Use wire between 0.5 mm^2 (20 AWG) and 1.5 mm^2 (16 AWG) for all signal wiring. Single twisted pair cable is readily available in 0.5 mm^2 (20 AWG).

CANbus termination

See also the section on EMC.

If your system has more than one CAN node, connect the nodes in a 'daisy chain' arrangement and terminate the connections of the two end nodes with a 120Ω resistor. If the end node is a HVLP-10/20, link CAN_L and CAN_TERM on the customer connector, a 120Ω resistor is built into the controller. If you have a single node system the termination resistor should be connected so that the bus operates correctly when configuration tools are used.

Signal connections

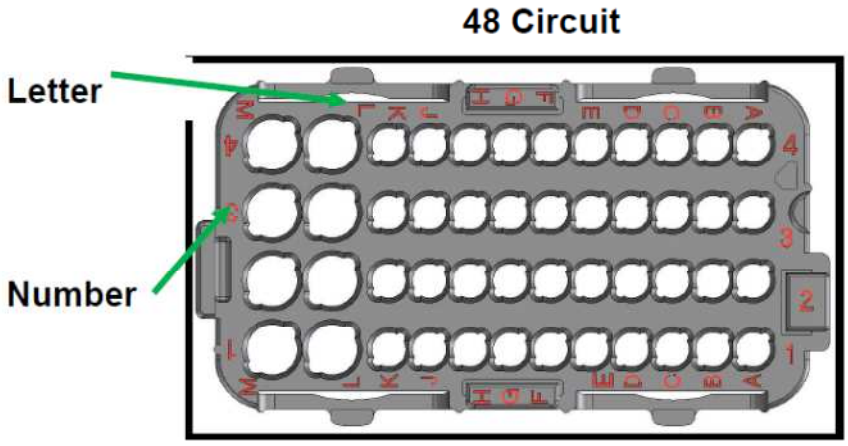
Signal connections are made to HVLP-10/20 via a 48 way Molex CMC connector.

Pins are numbered by rows 1 to 4 and columns A to M, for example B3.

There are 8 ways for 1.5mm^2 cable and 40 ways for 0.5mm^2 cable.

To maintain IP rating, unused positions must be sealed with appropriate hardware (available from Molex).

Please consult Molex document AS-64319-002 for details of how to use the connector and part numbers for the pins and plugs which are required.



*

Figure 16 - Customer Connector

The majority of pins are protected against short-circuits to the control logic supply positive or negative terminals.

Inserting contacts into connector housing pierces the sealing diaphragm to make the seal to the wire. To maintain IP rating, unused positions must be sealed with appropriate hardware (available from Tyco) if a contact is inserted and then subsequently removed.

The HVLP-10/20 functions of each pin are shown in Table 5.

⚠ Please note that the I/O pin allocations will likely change in units manufactured after September 2015. Please contact your Sevcon representative for further clarification.

Digital Input	Impedance to 12/24V supply	Impedance to Control 0V
1, 2	10k	12k
3, 4	12k	10k

Table 4 - Impedance at Digital Input Pins

Configure the digital input switches as active-high (switched to +24V) or active-low (switched to 0V). See section Digital inputs for more details.

When a switch is open the digital input pin sits at 0.5 x Vcc. The input sinks current in active-high configurations and sources current in active-low configurations.

Inserting contacts into connector housing pierces the sealing diaphragm to make the seal to the wire. To maintain IP rating, unused positions must be sealed with appropriate hardware (available from TE) if a contact is inserted and then subsequently removed.

Pin	Name	Type	What to connect	Maximum rating	Comment
A1	Resolver Cos Lo input	Analogue	Resolver type encoder	6V pk-pk from resolver	Shielded cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
B1	Resolver Cos input	Analogue	Resolver type encoder	6V pk-pk from resolver	Shielded cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
C1	Resolver Sin Lo input	Analogue	Resolver type enco	6V pk-pk from resolver	Shielded cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
D1	Resolver Sin input	Analogue	Resolver type encoder	6V pk-pk from resolver	Shielded cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
E1	Resolver excitation -	Analogue	Resolver type encoder	10kHz, 10V pk-pk (max) to encoder	Shielded cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
F1	Resolver excitation +	Analogue	Resolver type encoder	10kHz, 10V pk-pk (max) to encoder	Shielded cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.

Pin	Name	Type	What to connect	Maximum rating	Comment
G1	Active HVIL input 1	Digital	Interlock loop	0 > V > 36V	Not currently implemented Connect to terminal 1 (of 2) of the HVIL loop. There is no polarity requirement for this (i.e it can be connected either way round)
H1	Analogue 1 power supply +	Power	Programmable sensor supply output	I = 100 mA max	V = 0V to +10V, set in software
J1	Pot. 2 wiper in (AIN2_A)	Analogue	From potentiometer 2(A) wiper.	V = 9.5 V Zin = 50 kΩ	Suitable for potentiometers in the range 500 Ω to 10 kΩ, or voltage-output device (e.g. Sevcon linear accelerator) 0 to 5 V or 0 to 10 V. Ensure that at least 0.5V margin exists between the maximum valid throttle and the wire-off threshold
K1	Pot. 1 wiper in (AIN1_A)	Analogue	From potentiometer 1(A) wiper.	V = 9.5 V Zin = 50 kΩ	Suitable for potentiometers in the range 500 Ω to 10 kΩ, or voltage-output device (e.g. Sevcon linear accelerator) 0 to 5 V or 0 to 10 V. Ensure that at least 0.5V margin exists between the maximum valid throttle and the wire-off threshold
L1	12-24 Supply Input (KL30)	Power	Power Supply via suitable fuse	V = 14V (nominal) I = 1A (max)	This input supplies power from the low voltage source for all the logic circuits. The unit cannot operate without “Key switch in” supply. Referred to as KL30 in this table.
M1	12-24 Supply Input (KL30)	Power	Power Supply via suitable fuse	V = 14V or 24V (nominal) I = 1A (max)	This input supplies power from the low voltage source for all the logic circuits. The unit cannot operate without “Key switch in” supply. Referred to as KL30 in this table.

Pin	Name	Type	What to connect	Maximum rating	Comment
A2	Encoder B input	RS422 Differential	A/B type encoder	RS422 signal levels 120Ω differential input resistance	Not currently implemented. Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
B2	Encoder A input	RS422 Differential	A/B type encoder	RS422 signal levels 120Ω differential input resistance	Not currently implemented. Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
C2	Encoder W input	RS422 Differential	UVW type encoder	RS422 signal levels 120Ω differential input resistance	Not currently implemented. Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
D2	Encoder V input	RS422 Differential	UVW type encoder	RS422 signal levels 120Ω differential input resistance	Not currently implemented. Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
E2	Encoder U input	RS422 Differential	UVW type encoder	RS422 signal levels 120Ω differential input resistance	Not currently implemented. Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.

Pin	Name	Type	What to connect	Maximum rating	Comment
F2	Encoder power supply +	Power	To the positive supply input of the speed encoder	I = 200 mA	V = 0V to +10V, set in software
G2	SIN input	Analogue	SIN/COS quadrature type encoder	0-5V from encoder	Shielded cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
H2	Analogue input 3 (AIN3)	Analogue	To motor temperature thermistor (PTC type)	V = 13V I = 3mA Z_{out} = 4.7kΩ	
J2	Pot. 2 wiper in (AIN2_B)	Analogue	From potentiometer 2(B) wiper.	V = 9.5 V Zin = 50 kΩ	Suitable for potentiometers in the range 500 Ω to 10 kΩ, or voltage-output device (e.g. Sevcon linear accelerator) 0 to 5 V or 0 to 10 V. Ensure that at least 0.5V margin exists between the maximum valid throttle and the wire-off threshold
K2	Pot. 1 wiper in (AIN1_B)	Analogue	From potentiometer 1(B) wiper.	V = 9.5 V Zin = 50 kΩ	Suitable for potentiometers in the range 500 Ω to 10 kΩ, or voltage-output device (e.g. Sevcon linear accelerator) 0 to 5 V or 0 to 10 V. Ensure that at least 0.5V margin exists between the maximum valid throttle and the wire-off threshold
L2	Control 0V	Power	Logic power supply ground connection	0V	
M2	Key switch in (KL15)	Wake-Up	From key switch via suitable fuse	V = 12 or 24V (nominal) I = 4A max (Total of all contactor output currents)	This input supplies power for the contactor outputs (if used) as well as providing a wakeup signal to the controller The unit cannot operate without “Key switch in” supply. Referred to as KL15 in this table.

Pin	Name	Type	What to connect	Maximum rating	Comment
A3	Encoder ~B input	RS422 Differential	A/B type encoder	RS422 signal levels 120Ω differential input resistance	Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of AB, SIN/COS and resolver can be used at the same time.
B3	Encoder ~A input	RS422 Differential	A/B type encoder	RS422 signal levels 120Ω differential input resistance	Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
C3	Encoder ~W input	RS422 Differential	UVW type encoder	RS422 signal levels 120Ω differential input resistance	Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
D3	Encoder ~V input	RS422 Differential	UVW type encoder	RS422 signal levels 120Ω differential input resistance	Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
E3	Encoder ~U input	RS422 Differential	UVW type encoder	RS422 signal levels 120Ω differential input resistance	Shielded, twisted pair cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
F3	Encoder ground (power supply -ve)	Power	To the negative supply input (0 V) of the speed encoder	I = 200 mA V = 0.5 V	We recommend the use of shielded cable for the encoder wiring. Connect the shield to this pin only along with the negative supply.

Pin	Name	Type	What to connect	Maximum rating	Comment
G3	COS input	Analogue	SIN/COS quadrature type encoder	0-5V from encoder I = 10mA max	Shielded cables should be used for encoder connections. Controller must be set up correctly for the motor speed feedback type. Only one of UVW, AB, SIN/COS and resolver can be used at the same time.
H3	Digital Input 2	Digital	From digital switch input 2. This can be active high or active low	I = 2mA	
J3	Digital Input 1	Digital	From digital switch input 2. This can be active high or active low	I = 2mA	
K3	Active HVIL Source	Supply	Interlock loop	0 > V > 36V I = 10mA max	Source for HVIL (loop) if required. This output is a current limited isolated supply which can be used to originate the interlock system
L3	Digital High Side (Contactor) out 3	Out	To the switched low side of contactor or valve coil.	2.0A per output V = KL15	This output provides a high side switched output to the load
M3	Digital High Side (Contactor) out 4	Out	To the switched low side of contactor or valve coil.	2.0A per output V = KL15	This output provides a high side switched output to the load
A4	CAN Low	Comms	CANbus Low signal	V = 5 V	Maximum bus speed 1 Mbits/s.
B4	CAN Low	Comms	CANbus Low signal	V = 5 V	Maximum bus speed 1 Mbits/s.
C4	CAN termination	Comms	To terminate the CAN node link pin C4 to B4. This connects a 120Ω termination resistor, mounted inside the controller, across the CANbus.		Make the connection only if the controller is physically at the end of the CANbus network
D4	CAN High	Comms	CANbus High signal	V = 5 V	Maximum bus speed 1 Mbits/sec

Pin	Name	Type	What to connect	Maximum rating	Comment
E4	CAN High	Comms	CANbus High signal	V = 5 V	Maximum bus speed 1 Mbits/sec
F4	CAN ground	Power	To the ground of the external CAN bus		
G4	Active HVIL input 2	Digital	Interlock loop	0 > V > 36V I = 10mA max	Not currently implemented Connect to terminal 2 (of 2) of the HVIL loop. There is no polarity requirement for this (i.e it can be connected either way round)
H4	Digital Input 4	Digital	From digital switch input 4. This can be active high or active low	I = 2mA	
J4	Digital Input 3	Digital	From digital switch input 3. This can be active high or active low	I = 2mA	
K4	Active HVIL Sink	Supply	Interlock loop	0 > V > 36V I = 10mA max	Return signal for HVIL (loop) if required. This output is a current limited isolated supply which can be used to originate the interlock system
L4	Control 0V	Power	Logic power supply ground connection	0V	
M4	Control 0V	Power	Logic power supply ground connection	0V	

Table 5 I/O Connector pin out and wiring information

Control 0V Connections

The control 0V connections (L4, M4) should be connected to the vehicle or machine chassis as close the controller as possible.

SEVCON®

Chapter 4: Specification

Electrical

Input voltage – control supply

Nominal working voltage:	12V to 24V
Working voltage limits:	8V to 36V
Input protection:	Input protected against reverse connection of battery

Table 6 - Input voltage - control supply

Input voltage – DC Link supply

Nominal working voltage:	700V
Working voltage limits:	200V to 800V dc
Non-operational overvoltage limit:	850V dc
Input protection:	User must ensure that reverse polarity protection is provided external to the controller, for example by control of the main line contactor.
Surge protection	User must provide a pre-charge circuit to limit inrush current when the supply is first turned on
Ground fault detection	User must ensure that ground fault protection is provided external to the controller, for example by control of the main line contactor.

Table 7 - Input voltage - traction supply

Motor Output protection

Output current:	Reduced automatically from peak to continuous rating depending on the time a peak load is applied to the controller. Reduced automatically if operated outside normal temperature range.
Short-circuit:	Protected against any motor phase to B- or B+ at power-up. Protected against any motor phase to another motor phase at any time during operation. At switch-on HVLP-10/20 detects valid output loads are present before applying drive current.

Table 8 - Motor output protection



Repetitive short circuits may damage the controller.

Output ratings

The rating achievable in a particular application must be checked.

Parameter	HVLP10 (1m/sec)	HVLP10 (static) [1]	HVLP10 (Forced air) [2]	HVLP20 (Liquid cooled)
Continuous output current (A)	21	19	32	35[3]
Continuous output power @ 700V dc (kW)	13	11	20	22
Continuous output power @ 300V dc (kW)	5.5	4.5	8.5	9.5
Maximum output current (A) for 60s	27	24	48	53
Stall output current (A) for 5s	27	24	48	53

Table 9 - Output RatingsPower outputs in the above tables are approximate maximums. The actual power achieved will depend on modulation index, power factor and motor efficiency.

Notes:

1. *Static air operation requires convection over the heatsink, with the unit orientated vertically - see* Orientation *(page 3-2)*
2. *Forced air operation – currently untested, refer to Sevcon applications engineering*
3. *Depends on output cable (page x)*

CAN interface

CAN protocol:	H-protocol and I-protocol are supported J1939 compatible Physical layer uses ISO11898-2.
Baud rates supported:	1 Mbits/s (default), 500 kbits/s, 250 kbits/s, 125 kbits/s, 100 kbits/s and 50 kbits/s.

Table 10 - CAN interface

Control inputs and outputs

Digital inputs:	4 digital switch inputs (software configurable polarity). Active low inputs < 2V, active high inputs > Vb – 2 V Current 8mA when active
Analog inputs:	4 general purpose inputs which can be used for 2-wire potentiometers or dual throttle type devices: • 0 to 10V input • 10 bit resolution • Input resistance > 50kΩ All analog inputs can also be used as digital inputs.
Motor thermistor input	A motor thermistor (e.g. KTY84) can be connected between the analogue input and control OV • Internal excitation for the thermistor • When configured with a KTY84: – Accuracy ±5°C Measurement range will be -40°C to +200°C • Configurable to use a user defined characteristic.
Digital high side outputs	2 short circuit protected active high outputs. On / off control of the output only, no PWM voltage or current regulation Continuous sink current = 2A
Analogue supply outputs	2 short circuit protected supply outputs. Output voltage can be selected to 5V or 10V Output1: Programmable Vcc, 100mA max Output2: Encoder supply – 200mA max
Motor commutation sensor inputs:	Quadrature AB encoder signal inputs provided for control of induction motors UVW digital position sensor or sin-cos analogue position sensor or resolver inputs provided for control of permanent magnet motors AB and UVW are differential inputs with RS422 compliant signal specifications.
Resolver excitation	Two outputs, Exc+ and Exc-, provide a fixed 10kHz excitation to a speed feedback resolver. Each output is 4V max pk-pk, 100mA, protected against short-circuit. Consult the 'Resolver' section in the System Design section of this document for important information about resolver feedback.

Table 11 - Control inputs and outputs

Isolation

The HVLP comprises logic circuits which are connected to the logic circuits (12V or 24V) and the high voltage circuits connected to B+/B-.

As defined by ISO 6469-3, the logic circuits are class A, and the power circuits class B. There is a reinforced safety barrier between the power circuits and the logic circuits, and basic insulation between the power circuits and the enclosure. See Figure 17 below.

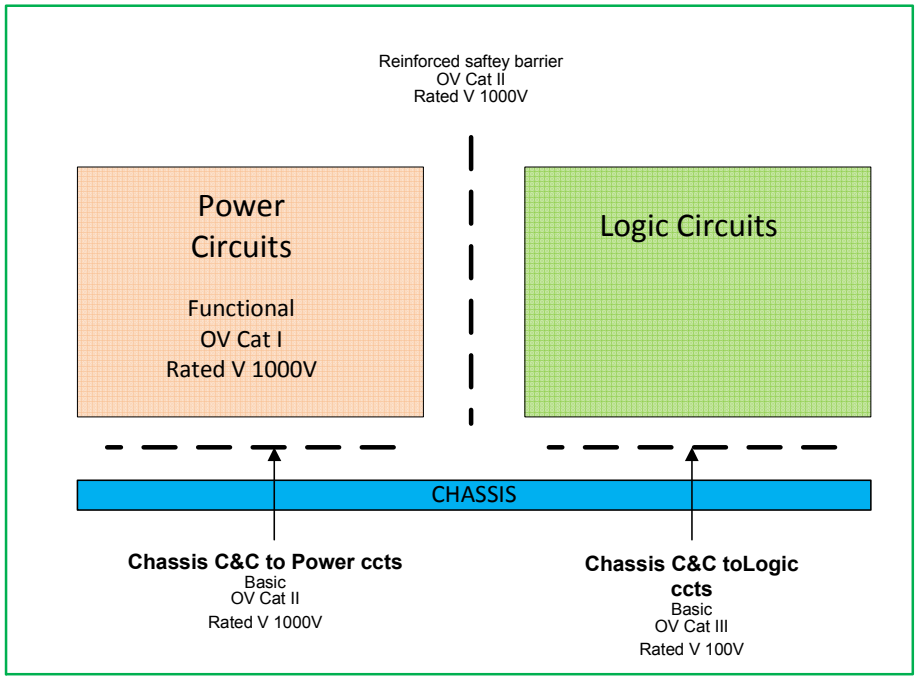


Figure 17 - Isolation Diagram

Power terminals to the case:	Withstands 2500 V ac[1]
Power terminals to control terminals	Withstands 3750 V ac[2]
Control terminals and CAN bus to the case	Withstands 500 V ac

Table 12 – Isolation

Notes:

1. Production test, isolation resistance >10MΩ
2. Type test only

Regulatory compliance

The controller is designed to meet -

Safety regulations	EN 60664 Insulation coordination for low voltage equipment ISO 6469 Electrically propelled vehicles Safety specifications UNECE Reg 100 Electric vehicles - Construction & safety ISO16230 (Draft) - Agricultural machinery and tractors. Safety of higher voltage electrical and electronic components and systems.
EMC regulations	DIRECTIVE 2009/64/EC OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 13 July 2009 on the suppression of radio interference produced by agricultural or forestry tractors (electromagnetic compatibility) UNECE reg 10 approval of vehicles with regard to electromagnetic compatibility

Table 13 - Regulatory compliance

Mechanical

Operating environment – HVLP-20 (Liquid Cooled)

Operating coolant temperature:	Liquid cooled; 50% water, 50% glycol Coolant temperature: -25°C to +65°C no current derating Coolant temperature: +65°C to +95°C and -40°C to -25°C with derating
Storage temperature range	-40°C to +100°C
Non-operation ambient temperature:	-10°C to +40°C (can be stored for up to 12 months in this ambient range)
Humidity:	95% at 40°C and 3% at 40°C
Ingress of dust and water:	Liquid cooled: IP67 and IP6K9K with connectors mated
Altitude	Designed to operate up to 2000m

Table 14 - Operating environment (HVLP-20)

Operating environment – HVLP-10 (Air Cooled)

Operating ambient temperature:	Air cooled: -30°C to +45°C no current derating +45°C to +85°C and -40°C to -30°C with derating
Non-operation ambient temperature:	Air cooled: -40°C to +85°C (can be stored for up to 12 months in this ambient range)
Humidity:	95% at 40°C and 3% at 40°C
Ingress of dust and water:	Air cooled: IP67 and IP6K9K with connectors mated

Table 15 - Operating environment (HVLP-10)

Shock and vibration

Thermal shock	EN60068-2-14, Test Na
Repetitive shock:	EN 60068-2-27, Test Ea 50 g peak 3 orthogonal axes, 3+ and 3– in each axis, 11 ms pulse width
Drop test:	EN 60068-2-32:1993 Test Ed: Free fall, appendix B, Table 1
Bump:	40 g peak, 6 ms, 1000 bumps in each direction repetition rate 1 to 3 Hz.
Vibration:	3 g, 5 Hz to 500 Hz
Random vibration:	20 Hz to 500 Hz, acceleration spectral density 0.05 g2/Hz (equivalent to 4.9 grms)

Table 16 - Shock and Vibration

Weight

Controller weight

HVLP-10 Air Cooled	3.7kg
HVLP-20 Liquid Cooled	2.3kg (dry weight)

Table 17 - Weight

Dimensions HVLP-10/20

HVLP-10 (Air Cooled):

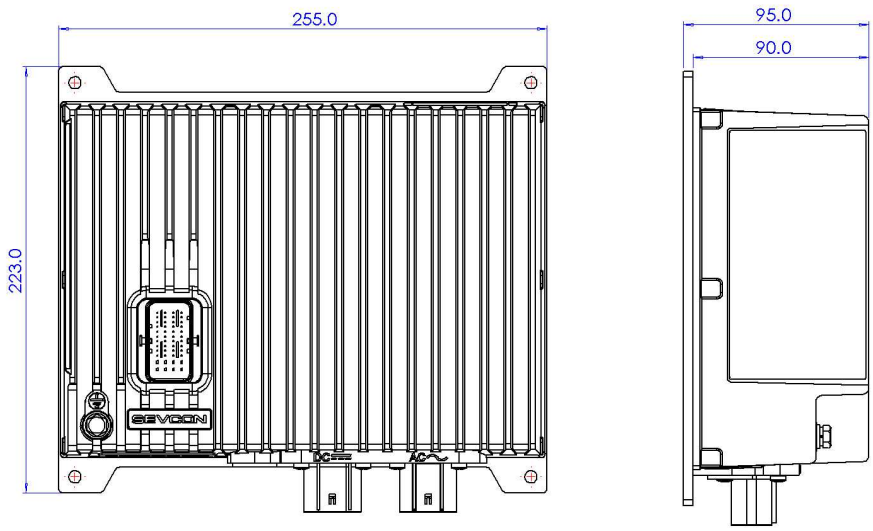


Figure 18 - HVLP-10 Dimensions

HVLP-20 (Liquid Cooled)

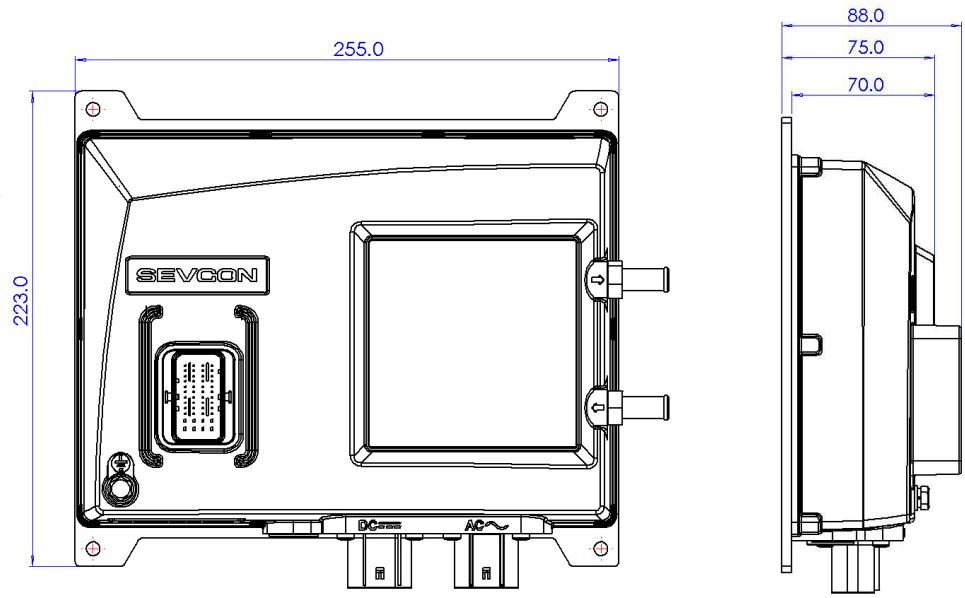


Figure 19 - HVLP-20 Dimensions

(Note 1 Above dimensions expected to be within +/-2.0mm)

(Note 2 Dimensions illustrated do not include allowance for connectors, cables and hoses)

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Chapter 5:
System design

Selecting the HVLP-10/20 model

Matching motor and controller ratings is not an exact exercise and therefore you may need to perform iterative calculations. The main considerations when choosing an appropriate HVLP-10/20 controller are described below.

Circuit configuration

A basic single traction configuration (Figure 20) is provided as a starting point for new designs. Given the flexibility of the I/O it is possible to configure a wide range of systems.

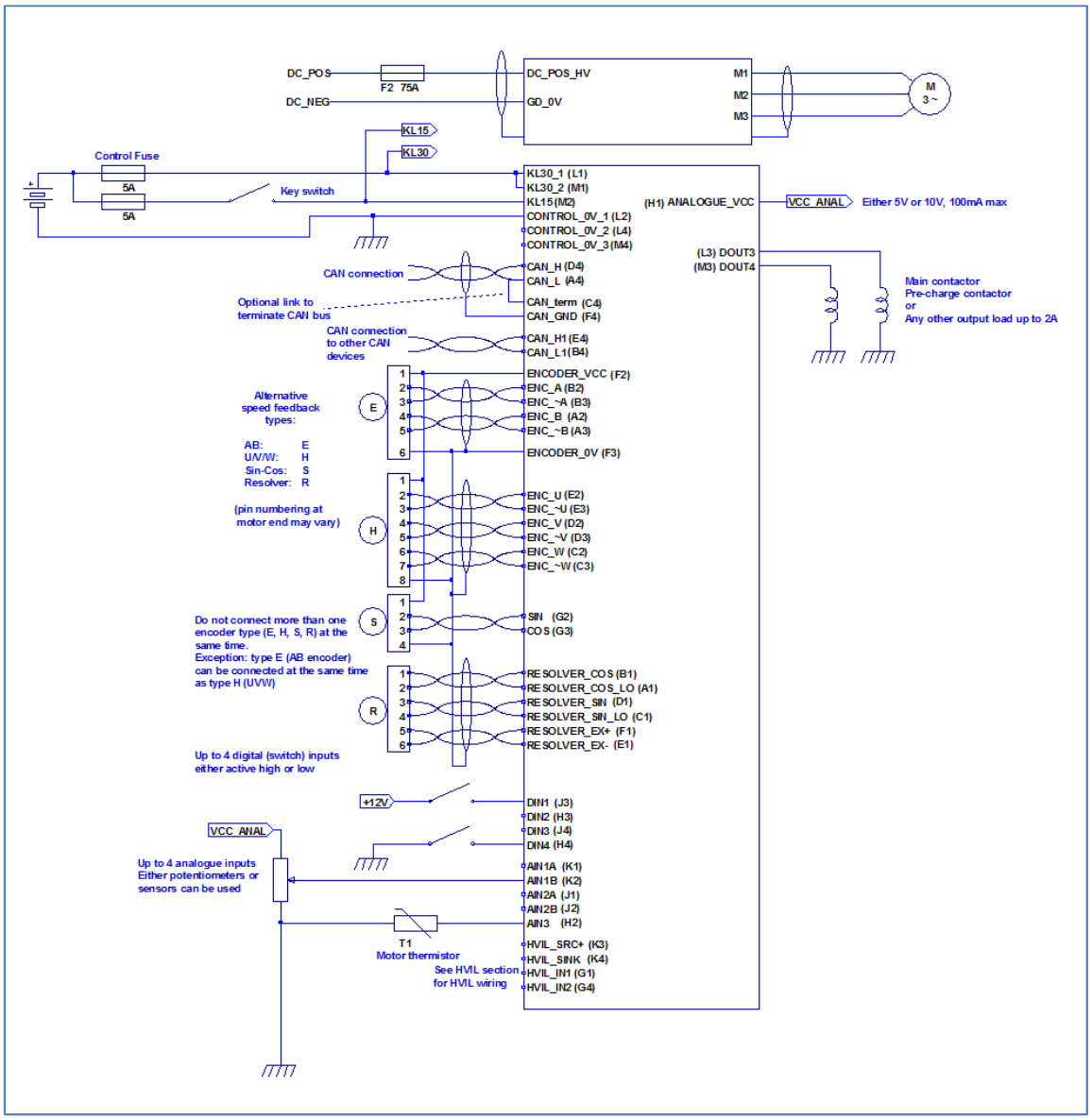


Figure 20 - HVLP Wiring Diagram

Auxiliary Components (including fuses)

Main Contactor and Precharge circuit

HVLP-10/20 does not offer any internal pre-charge function. Pre-charge must be done externally to the HVLP unit. An external device must be used to isolate the HVLP-10/20 from the vehicle battery. This external device is also responsible for any capacitor precharging required to prevent damage to the line contactor tips.

It is recommended that the precharge circuit is connected as shown in Figure 21 .The circuit is intended to operate in the following manner:-

- At start-up K1 must be energised, the controller DC bus voltage should be monitored and should have reached 90% of the battery voltage within 5s.
- If the dc bus voltage is more than 90% of battery voltage the main contactor K2 can now be closed. If the dc bus voltage is too low this indicates a fault and K1 should be de-energised.

At shut-down K1 and K2 may be de-energised simultaneously.

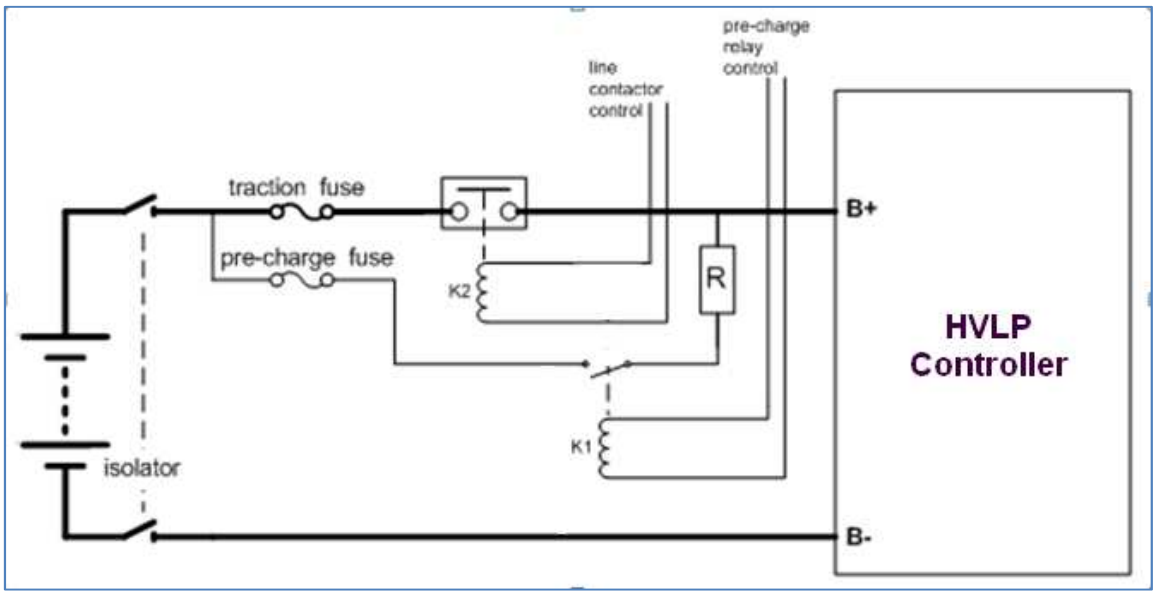


Figure 21 - Line contactor and pre-charge wiring diagram

Components required for the precharge and discharge circuit are listed in Table 18.

Name	Function	Required specification	Comment
F1	Pre-charge fuse	10A/20A 700V/1000V dc Slow blow preferred	e.g. Cooper Bussman 170M1308 (700V rating) Cooper Bussman 170M4802(1000V rating, 20A)
F2	Main fuse	50A 700V/1000V dc Semiconductor fuse	e.g. Cooper Bussman 170M1314 (700V rating) Cooper Bussman 170M4807(1000V rating)
K1	Pre-charge contactor/relay	Single pole 700V/1000V dc 10A	
K2	Main line contactor	Single pole 900V dc 50A	Must be interlocked so that K2 can only close if K1 is energised.
R1	Pre-charge and discharge resistor	Resistance 220Ω to 1kΩ Pulse voltage 1000V Energy pulse 150J Power 25W	Must be overload protected with thermal cut out and or fuse

Table 18 - Pre-charge Components

Failure to use a capacitor precharge circuit may lead to damage to the controller. Note that the dc bus capacitors remain charged at hazardous voltages for several minutes.

Note: 700V fuses suggested above should only be used if the nominal DCbus voltage is 700VDC or less, otherwise 1000V fuses should be used.

Digital Outputs from HVLP-10/20

The digital outputs can provide up to 2A each and can be used for various purposes (e.g. lamps, relays, contactors). But users should be aware that these outputs are powered from KL15 (key-on supply). If the key-switch is turned off the contactors would open immediately even if the motor was turning.

See Figure 22 for a circuit diagram showing the internal components.

The outputs are short-circuit proof and thermally protected, but users should be aware that if the flyback diode is connected in parallel with the load inductor, the controller will be damaged in the event of reverse battery.

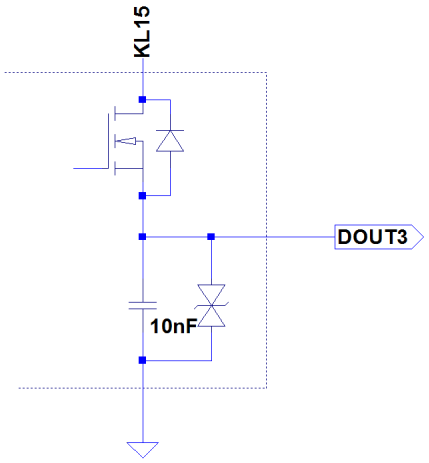


Figure 22 - Digital Output Schematic

Note the following:

- The digital output is ON/OFF only
- The load should be referred to the logic ground
- The load must not have a parallel diode, as this would violate the reverse battery protection

Motor speed sensor (AB encoder)

A 6-wire connection is provided for open-collector quadrature pulse encoder devices. These types of encoder are optimized for accurate speed measurement, required for efficient control of induction motors.

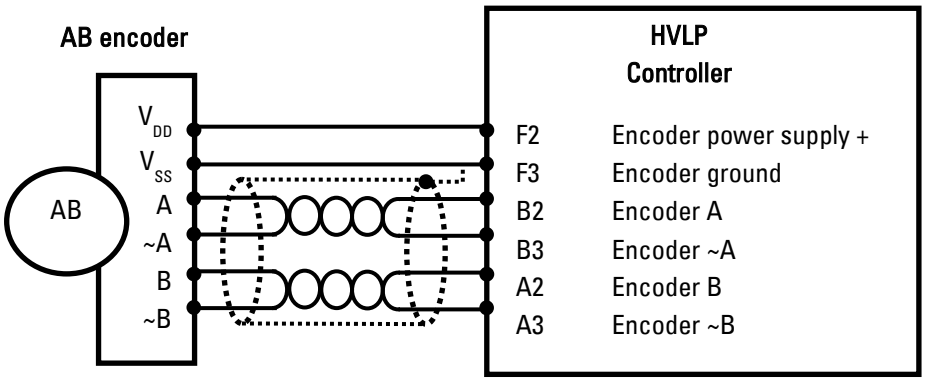


Figure 23 - Sample wiring for an AB quadrature speed encoder (encoder pin names may vary).

The number of encoder pulses per revolutions (**n**) and the maximum motor speed (**N**) are related to, and limited by, the maximum frequency of the encoder signal ($\mathbf{f}_{max}$).

$$f_{max}(Hz) = \frac{n(per\ rev) \times N(rpm)}{60}$$

with f_{max} limited to 200 kHz, and

$$N_{max}(rpm) = \frac{20000(rpm)}{(p/2)}$$

Please refer to page 6-11 for information on entering encoder settings.

Motor commutation sensor

Commutation sensors are designed to measure the absolute position of the rotor shaft within the motor, rather than its rotational speed. Rotor position information is used for control of permanent magnet motors, as it allows the controller to energise the motor phases appropriately based on the measured position of the magnets on the rotor.

UVW Commutation Sensors

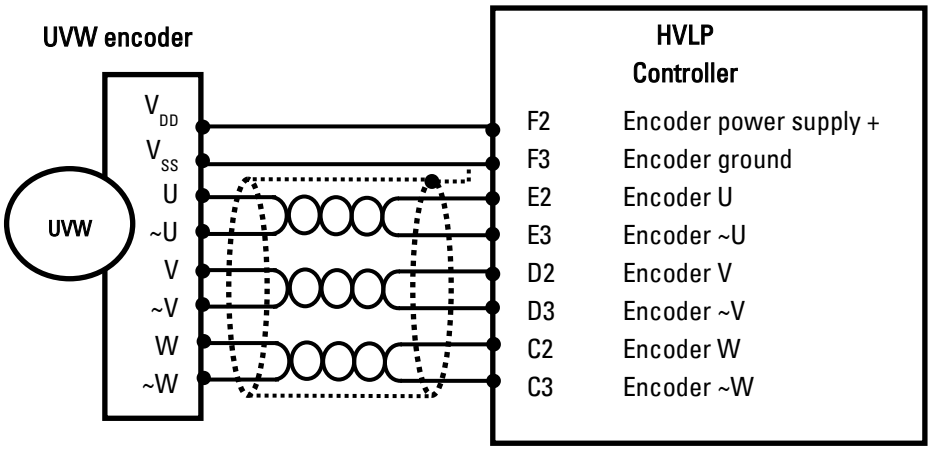


Figure 24 - Sample wiring for a UVW commutation sensor (encoder pin names may vary).

6 digital inputs are provided for differential UVW encoders, with a 5-10V power supply. The encoder should provide one pulse on each UVW channel per electrical cycle of the motor, and each pulse should be 120° out of phase with the others and have a 50% duty cycle. The RS422 inverse of these signals must also be present on the ~U,~V,~W inputs.

Figure 25 shows an example pulse train for the U, V and W inputs.

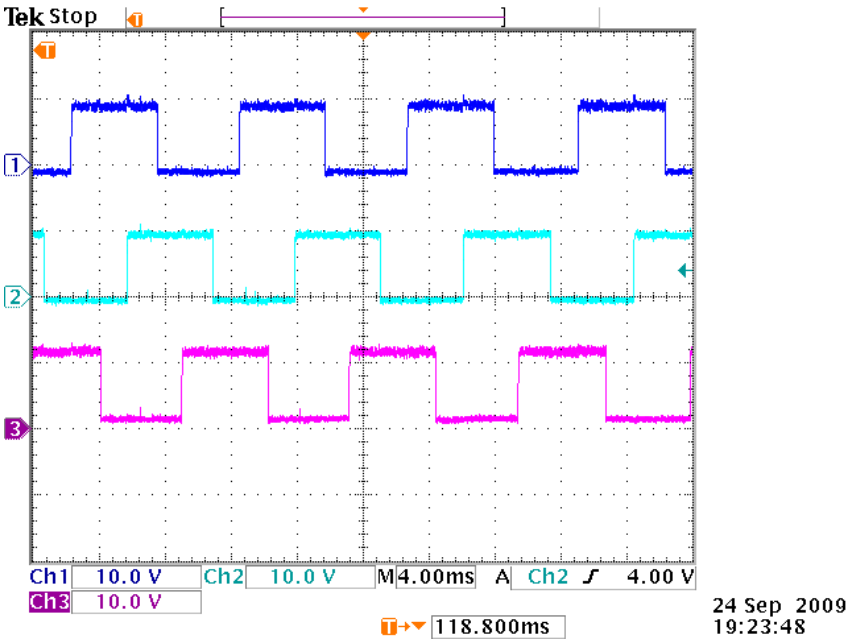


Figure 25 - Example pulse train from a UVW commutation sensor

Sin-Cos Commutation Sensor

Analogue sin-cos encoders generate a number of sin and cosine waves per mechanical rotation of the motor.

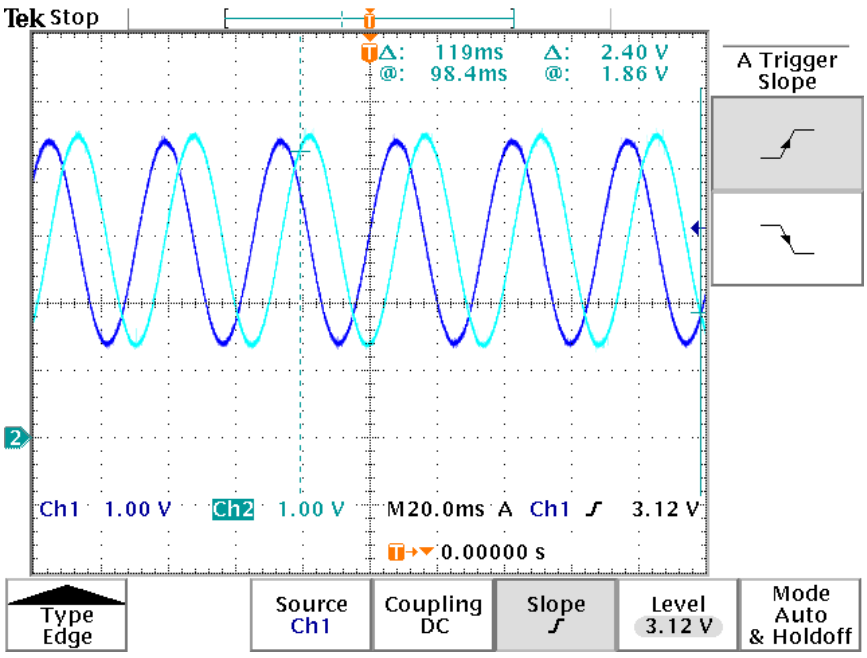


Figure 26 - Example of signals from a sin-cos position sensor

The controller is able to control motors with sin-cos sensors that produce multiple sin and cosine waves per mechanical rotation. However, it is required that the number of pole pairs in the motor is an integer multiple of the number of sin-cos waves per rotation (itself an integer only).

For examples:

- An encoder that produces 1 wave per rotation can be used with motors with 1 pole pair, 2 pole pairs, 3 pole pairs, etc…
- An encoder that produces 3 waves per rotation can be used with motors with 3 pole pairs, 6 pole pairs, 9 pole pairs, etc…
- An encoder that produces 5 waves per rotation can be used with motors with 5 pole pairs, 10 pole pairs, 15 pole pairs, etc...

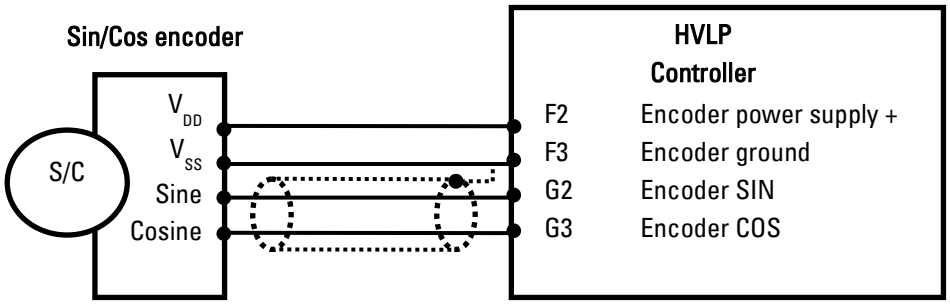


Figure 27 - Sample wiring for a sin-cos commutation sensor

Sin-cos encoders are typically powered by a 5V supply. Therefore it is important to ensure that the controller is configured to supply 5V on Encoder Vcc. This should be done by setting the encoder configuration object dictionary entry at 4630_h.

Resolver

The resolver is a position feedback device which requires a sinusoidal excitation (provided by the HVLP-10/20 controller). The speed feedback consists of two pairs of signals: a Resolver Sin pair and a Resolver Cos pair (Figure 28). These signals consist of the excitation frequency modulated by the sin or cosine (respectively) of the rotor position.

The following issues are important when selecting a resolver: mechanical, environmental and electrical specification.

- Mechanical specification includes e.g. material; dimensions; max speed; max angular acceleration; rotor moment of inertia; resistance to shock; resistance to vibration; permitted axial offset; permitted radial runout (deviation from true circle) of motor shaft.
- Environmental specification includes: max operating temperature (including self-heating); IP rating.
- Electrical specification includes: number of pole pairs (p); transformer ratio (Rt); angular error tolerance; residual voltage; recommended input voltage; recommended operating frequency; maximum primary current.

Regarding mechanical and environmental specification, the user must select a resolver appropriate to the environment of the application.

Regarding electrical specification, the following features are required of a resolver to work with the HVLP-10/20:

- Transformer ratio = 0.2 to 0.6
- Excitation frequency = 10kHz
- Excitation voltage = 5Vrms nominal (a resolver specified to 5Vrms or higher will be acceptable)

The controller is able to control motors with multiple pole-pair resolvers (i.e. returning multiple sin and cosine waves per mechanical rotation). However, it is required that the number of pole pairs in the motor is an integer multiple of the number of resolver pole pairs.

For example:

- A single pole pair encoder can be used with motors that have 1 pole pair, 2 pole pairs, 3 pole pairs, etc...
- A 3 pole pair encoder can be used with motors that have 3 pole pairs, 6 pole pairs, 9 pole pairs, etc...
- A 5 pole pair encoder can be used with motors that have 5 pole pairs, 10 pole pairs, 15 pole pairs, etc...

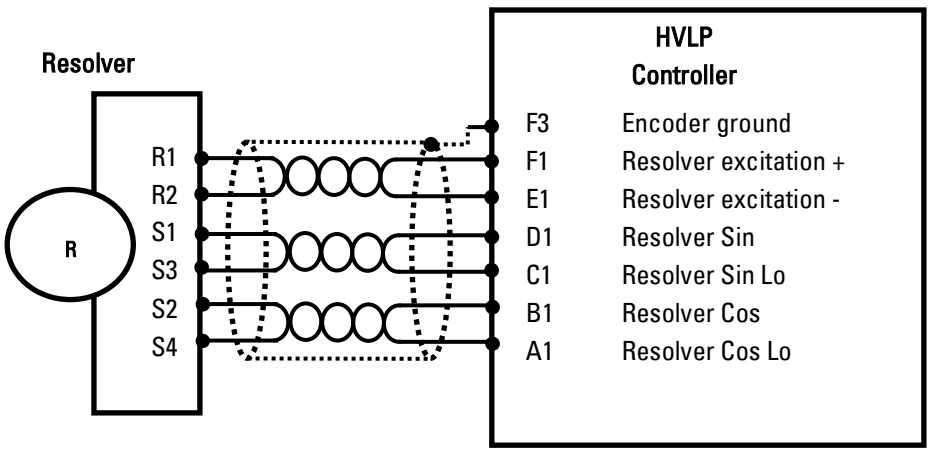


Figure 28 -Sample wiring for a resolver

Only connect a single type of encoder to the HVLP-10/20 at any given time.

Initial power up sequence

Incorrectly wired or configured vehicles may behave in unexpected ways.

Checks prior to power up

Follow this checklist prior to applying power to your system:

- Confirm all connections are tightened to specified level.
- Ensure all plugs are fully inserted.
- Confirm power wiring connections are made to the correct terminals (B+, B-, +, M1, M2 and M3).
- Ensure the controller is securely mounted (from a mechanical and thermal perspective).
- Ensure there is adequate coolant flow.
- Check the routing of cables is safe with no risk of short circuit, overheating or cable insulation wear due to rubbing.

Checks after power is applied

Apply power and do the following:

- Use Sevcon DVT (see page 6-2) or any configuration tool to complete the configuration process which starts on page 6-9.
- Using the drive controls ensure the motor rotates in the expected direction. If they do not, check the motor wiring, encoder wiring and encoder configuration (page 6-11).

Discharge sequence after power down

Hazardous voltages will remain on the controller internally and on exposed power terminals for a period of time after the main power connections have been removed.

Hazardous voltages may remain on the controller internally and on exposed power terminals after the main battery power connections and keyswitch power supplies have been removed if the controller is connected to a rotating permanent magnet motor.

Controller discharge profiles

The following graph shows the standard controller discharge curve which should be observed. This curve only applies if the discharge circuit in Main Contactor and Precharge circuit is not used or has been disconnected.

Do not open the unit or work near exposed power terminals until the voltage has reduced to a safe level.

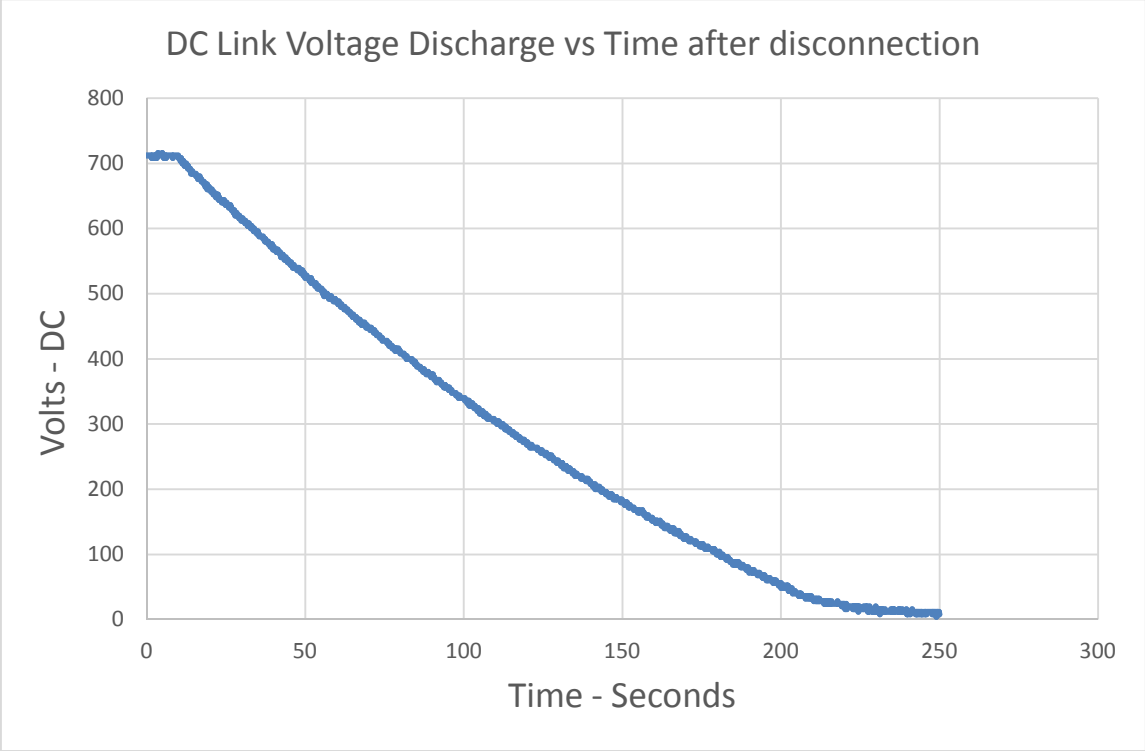


Figure 29 – Typical discharge curve for the HVLP-10/20 controller

To determine the required time, note the time at which the “Voltage (maximum)” curve crosses the horizontal gridline corresponding to the supply voltage. Then note the time at which it crosses the horizontal gridline corresponding to the safe voltage level (normally considered to be 60V). The difference between the two times is the safe discharge time period.

This information is for guidance only, and before any work is carried out on the high voltage connections, it is essential the voltage is checked using a DVM and probes rated for at least 1000V



To achieve a shorter discharge time the vehicle system designer must ensure that the line contactor and capacitor precharge circuit also incorporates a discharge circuit which is activated once the main line contactor is opened.



If the controller is connected to a permanent magnet motor and the motor is rotating the discharge circuit will be exposed to the rectified back emf even if the controller keyswitch power supply is turned off.

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Chapter 6: Configuration

Introduction

This section covers what you need to do to configure HVLP-10/20's software once you have designed and installed your hardware. All of HVLP-10/20's parameters have a default value and the amount of configuration needed is dependent on your particular system.

The main topics are:

- DVT configuration tool: installation and use
- CANopen: an introduction to the protocol and its use in Sevcon products
- An overview of the configuration process outlining what needs to be done and the order in which it must be done
- The configuration steps

DVT configuration tool

DVT is Sevcon's proprietary configuration tool. It allows the user to monitor, configure and duplicate the parameters of any Sevcon CANopen node such as the HVLP-10/20 controller. The information presented here is an overview only. Contact Sevcon for more information about DVT and the functions it provides.

DVT functionality

DVT provides the following facilities:

- Configuration of controller IO, CANbus and motor parameters
- Loading DCF configuration file into the Sevcon controller
- Saving of DCF configuration file from controller to file on computer
- Controller status and fault diagnosis
- Data logging of controller/motor performance on test bench or vehicle
- Update controller firmware

Saving, duplicating and restoring a node's configuration

You can use DVT to:

- Save a node's configuration. This can be used at some later date to clone the node's configuration.
- Duplicate a node's configuration, in real time, to another node on the CANbus.
- Restore a configuration to a node.

For example, if you want to save the HVLP-10/20 controller configuration, you will need to create a DCF file. To do this, open the helper by clicking the  icon at the top of the DVT main window.

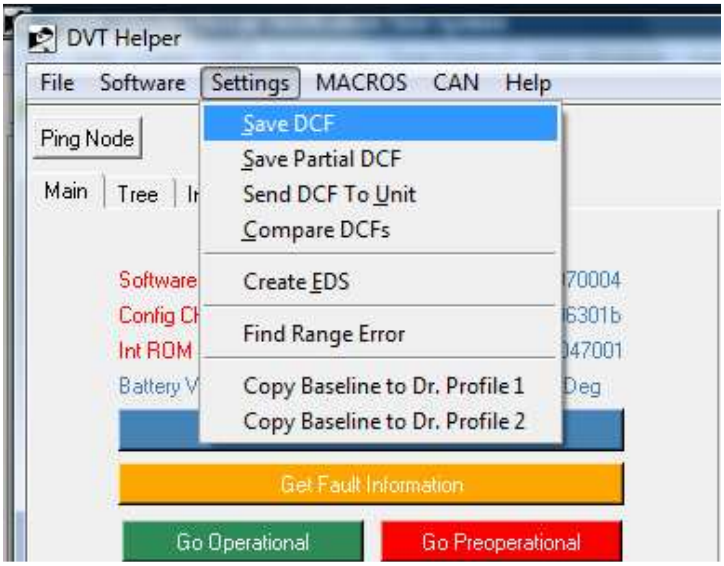


Figure 30 - DVT Helper

Data Logging

You can use DVT to monitor data or parameters of a Sevcon node in real time and graph the data.

CANopen Configuration

This section assumes you have an understanding of CAN and are familiar with its use. If you are new to CAN or CANopen please refer to the CiA (CAN in Automation) website, www.can-cia.org for further information.

The following information provides an introduction to the important CANopen terminology used in this manual and how it relates to the configuration of your HVLP-10/20 controller.

CANopen protocol

CANopen is a CAN higher layer protocol and is defined in the DS301 'Application Layer and Communication Profile' specification. All CANopen devices must adhere to this standard. To provide greater standardization and interoperability with 3rd party devices.

Object Dictionary

The HVLP-10/20 controller uses CANopen principally for configuration purposes. All settings and configuration can be accessed via the controller's object dictionary, and parameters can be monitored and adjusted using CANopen's SDO protocol. This protocol

is also implemented in DVT, allowing it to be used as the primary device configuration tool.

Any device connected to the CANopen network is entirely described by its Object Dictionary. The object dictionary defines the interface to a device. Within the Object Dictionary, all parameters are assigned a unique index and sub-index. Thus, all parameters can be uniquely addressed by this pair of numbers.

There are two important text files associated with the Object Dictionary. These are:

EDS (Electronic Data Sheet)

An EDS is a text file representation of the Object Dictionary structure only. It contains no data values. The EDS is used by configuration software such as Sevcon's DVT to describe the structure of a node's Object Dictionary. An EDS for each HVLP-10/20 model and software version, is available from Sevcon. The EDS file format is described in the DSP306 – Electronic Data Sheet Specification.



Each Object Dictionary matches a particular HVLP-10/20 software revision, and its structure is hard coded into the controller software.

DCF (Device Configuration File)

This is a text file similar to an EDS except that it contains data values as well as the Object Dictionary structure.

DCFs are used to:

- Download a complete pre-defined configuration to a node's Object Dictionary.
- Save the current configuration of a node's Object Dictionary for future use.

Network Configuration

General

Use the following procedure to setup the HVLP-10/20 controller on the network:

1. Set node ID and baud rate in 5900h to the required values. Node IDs must be unique, and the baud rate must be the same for each node.
2. Set SYNC COB-ID in 1005_h to 0x40000080 for the SYNC producer, or to 0x00000080 for the SYNC consumers. Bit 30 is set to indicate to a node if it is the SYNC producer. Only one node in the network should be configured as the SYNC producer. This should normally be the VCU. On the SYNC producer, set the SYNC rate in $1006_{h.}$
3. Set the EMCY message COB-ID to either 0x00000000 or 0x80 + node ID in 1014_h. Setting to 0x00000000 will automatically calculate the EMCY COBID from the node ID.



EMCY COB-IDs must be configured correctly to ensure the master handles EMCYs from slaves correctly.

4. Configure the heartbeat producer rate in 1017_h. This is the rate at which this node will transmit heartbeat messages.
5. Configure the heartbeat consumer rate in 1016_h. A consumer should be configured for each node to be monitored.

Heartbeats must be configured correctly for correct network error handling. The master node should monitor heartbeats from all slave nodes. Slave nodes should, at a minimum, monitor heartbeats from the master node.

Loss of CANbus communication from any one node must cause a heartbeat fault to occur.

6. Where supported, hardware CANbus fault detection should be enabled at 5901_h to ensure CANbus fault detection is always enabled.
7. Configure additional SDO servers. An SDO server allows another CANopen device to SDO read/write from a node's object dictionary. Each node has one default SDO server (1200_h) which is reserved for communication with configuration tools like DVT or the calibrator. Another 3 SDO servers can be configured at 1201_h to 1203_h. These should be used as follows:
    a. On slave nodes, configure a server to allow the master node to communicate.
    b. If there is a display in the system, configure a server to allow the display access.
8. Configure RPDOs (1400_h to $17FF_h$) and TPDOs (1800_h to $1BFF_h$) appropriately for the system.
9. Configure the RPDO timeout function if required. See section PDO (page 6-5) for more information.
10. Set up the VCU to comply with the required H-protocol and I-protocol messages for motor control and I/O control.

3rd Party CANopen Devices

At power up, the HVLP master will communicate with all slave nodes to identify which nodes are Sevcon devices and which are not using the vendor ID in 1018_h. This instructs the HVLP how to handle EMCY messages from each node.

HVLP knows how to react to EMCYs (faults) from Sevcon slaves and can take appropriate action. HVLP does not know how to react to EMCYs from 3rd party devices, so the required fault reaction to 3rd party device EMCYs must be set at 2830_h.

PDOs

The controller supports 5 RPDOs (receive PDOs) and 5 TPDOs (transmit PDOs). Up to 8 Object Dictionary entries can be mapped to each PDO. Every PDO must have a unique identifier (COB-ID).

Setup RPDOs and TPDOs to transmit and receive non-critical data between CANbus nodes.

H-protocol and I-protocol should be used to communicate motor and I/O data.

The easiest way to do this is using DVT. If you are using a 3rd party configuration tool, the relevant Object Dictionary indices are listed in Table 19.

Feature	Object indices	Notes
Input mapping	1400h-15FFh	RPDO communication parameters
	1600h-17FFh	RPDO mapping
Output mapping	1800h-19FFh	TPDO communication parameters
	1A00h-1BFFh	TPDO mapping

Table 19 Objects associated with mapping

HVLP-10/20 supports RPDO timeout fault detection. This can set a warning, drive inhibit or severe fault depending on the configuration in 5902_h.

RPDO timeout can be used for non-CANopen systems which do not support heartbeating. By default, RPDO timeout is disabled, and normal CANopen heartbeating protocol (see section Network Configuration (page 6-4)) is assumed to be used.

H-Protocol

Although device configuration takes place using standard CANopen messages which are used to access the device's Object Dictionary, the proprietary H-Protocol is used to communicate motor control and status information over a CAN network.

H-Protocol is an enhanced motor control protocol. It is designed specifically for use with traction applications that require fast response times and a fine degree of control, but is equally suited to any electronic motor control application.

H-Protocol allows a VCU to control:

- Inverter bridge state
- Motor output torque
- Motor speed limits in both the forward and reverse directions
- DC current limits for both drive and regeneration
- Overall torque limits for both driving and braking torques

H-Protocol allows a VCU to monitor:

- Motor speed and actual output torque
- Motor, inverter, and internal component temperatures
- DC voltages and currents
- Inverter bridge state
- Fault codes

J1939 Compatibility

The network layer used by H-Protocol is intended to be compatible with J1939 type networks. Whilst non-standard PGNs are used to represent the data, the J1939 concepts for routing data and providing peer-to-peer communication between a VCU and motor controller are used heavily by H-Protocol.

The PGNs used by H-Protocol can be configured by adjusting the parameters in the object dictionary at index 3F00h. Any PDU1 format PGNs can be used by H-Protocol. However, the following default configuration is recommended wherever possible:

Name	Description	PGN (dec)	PGN (hex)
HC1	Torque demand message 1	69632	0x11000
HC2	Torque demand message 2	69888	0x11100
HC3	Battery demands	70144	0x11200
HS1	Torque feedback	71680	0x11800
HS2	Inverter status feedback	71936	0x11900
HS3	Temperature feedback	72192	0x11a00
HS4	Fault Information	72448	0x11b00
HS5	Encoder Information	72704	0x11c00
HD1	Debug Currents	75776	0x12800
HD2	Debug Voltages	76032	0x12900
HD3	Debug Temperatures A	76288	0x12a00
HD4	Debug Temperatures B	76544	0x12b00
HD5	Debug Control PMAC	76800	0x12c00
HD6	Debug Control FV	77056	0x12d00

Further details regarding the layout of signals within these messages is available from Sevcon on request.

If a VCU is not available to generate motor control demands in H-Protocol format, TracApp can be used instead.

H-Protocol includes an integrated timeout feature. If the motor controller does not receive H-Protocol commands for a specified timeout period, the motor will automatically output zero torque, and will also attempt to report this condition to the controlling device.

I-Protocol

I-Protocol works alongside H-Protocol to allow monitoring and provide control of digital and analogue input and output devices.

I-Protocol allows a VCU to control:

- Digital outputs
- Analogue outputs (contactor drives)

I-Protocol allows a VCU to monitor:

- State of digital inputs
- Analogue input voltages
- Fault codes

J1939 Compatibility

Like H-Protocol, I-Protocol is intended to be compatible with J1939, and communicates using PDU1 type PGNs. The PGNs used by the IO controller can be configured in the Object Dictionary at index 3F01h. However, the default PGNs listed below should be used if possible:

Name	Description	PGN (dec)	PGN (hex)
UO1	User Analogue Output Control 1-4	81920	0x14000
UO2	User Analogue Output Control 5-8	82176	0x14100
UF1	User I/O Fault Status and Digital Inputs	83968	0x14800
UI1	User Analogue Inputs 1-4	84224	0x14900
UI2	User Analogue Inputs 5-8	84480	0x14a00
UI3	User Analogue Inputs 9-12	84736	0x14b00
UI4	User Analogue Inputs 13-16	84992	0x14c00

Further details regarding the layout of signals within these messages is available from Sevcon on request.

If a VCU is not available to generate motor control demands in I-Protocol format, TracApp can be used instead.

Configuration process overview

Electric vehicles can be dangerous. All testing, fault-finding and adjustment should be carried out by competent personnel.

We recommend saving parameter values by creating a DCF, before making any alterations so you can refer to, or restore the default values if necessary. Do this using DVT.

This part of the manual assumes you have a vehicle designed and correctly wired up with a CANopen network setup. Before you can safely drive your vehicle it is necessary to go through the following process in the order presented:

Step	Stage	Page
1	Motor characterisation	A-6-10

Table 20 - Configuration requirements

Access authorization

To prevent unauthorized changes to the controller configuration there are 4 levels of accessibility: (1) User, (2) Service Engineer, (3) Dealer and (4) OEM Engineering. The lowest level is (1), allowing read only access, and the highest level is (4) allowing authorization to change any parameter.

To login write your 32-bit password to object 5000_h sub-index 4. The access level can be read back from sub-index 1. The passwords for access levels 1 to 4 are fully configurable by the customer in 5000_h sub-indices 5-8 so they can be made unique. Log in at access level 4 is required to change the passwords.

NOTE: Make a note of any changed passwords. Resetting forgotten passwords can only be performed by SEVCON.

How NMT state affects access to parameters

Some important objects can only be written to when the controller is in the pre-operational state. DVT takes HVLP-10/20 in and out of this state as required.

If you are not using DVT you may need to request the CANopen network to enter pre-operational before all objects can be written to.

The NMT state can be determined by monitoring the heartbeats or by reading 5110_h. 05 = operational and 7F = pre-operational.

Motor characterisation

Ensure you have completed the CANopen network setup process.

Determining Motor Parameters

Key motor parameters for permanent magnet ac and induction motor are adjustable in 0x4641.



Contact Sevcon for more information on how to set up the HVLP-10/20 controller for supported motor types.

Speed Limit Tuning

The speed limit is controlled using PI loops, which can be configured in the Object Dictionary at index 4651h. The following parameters can be configured:

- Standard proportional and integral gains (4651h, 1+2). Used to configure the loops during normal operation.
- Low speed proportional and integral gains (4651h, 3+5). Used to configure the loops at low speeds (<50 RPM) and during hill hold. These are normally set lower than the standard gains to dampen oscillation as the vehicle comes to a stop.
- Roll back integral gain (4651h, 4). Used to boost the integral term to prevent vehicle roll-off down inclines, particularly when Hill Hold is enabled. Normally, this gain is higher than the standard integral gain.
- dw/dt gain (4651h, 6). For speed mode this is used to boost the torque output in speed mode, when a large increase in speed demand occurs. For torque mode this is used to apply compensatory torque to damped oscillations induced from the vehicle drive train. Not used in torque mode.
- Integral initialization factor (4651h, 7). Used to initialize the integral term on entry to speed limit in torque mode. This factor is multiplied by the actual torque to set the integral term. Not used in speed mode.
- Speed filter pole (4651h, 8). Used to filter speed used by speed limit loop.
- Speed pre-limit threshold (4651h, 9). Speed limit algorithm will start cutting back torque when speed is within this threshold of the maximum.

I/O configuration

Ensure you have completed the CANopen network setup and Motor Characterization processes described above.

The individual characteristics of the I/O in your application need to be setup. Setup each hardware I/O object, including wire-off protection.

I/O control is managed using the I-Protocol. Refer to the I-Protocol documentation for more information.

Before starting the mapping process it is a good idea to draw out a map of what you want to do. The amount of mapping required depends on the electrical wiring of your vehicle. Check to see if the default settings satisfy your needs before making changes.

Encoder

It is important that the encoder parameters are entered correctly. If this information is not correct, motor control may not be possible.

General encoder settings

Encoder selection and settings applicable to all encoder types are contained in 4630_h and 4631_h.

HVLP supports four encoder types,

Resolver absolute position sensor
Sine-Cosine absolute position sensor
UVW absolute position sensor
AB incremental speed encoder

AB is not suitable for synchronous motors (e.g. permanent magnet AC).

- For sensors requiring a DC power supply (Sine-Cosine, UVW, AB), set the required sensor supply voltage (from 5V to 10V) in 4630_h.
- If the encoder sequence is incorrect with respect to the motor, then this inversion can be performed within 4631_h, without rewiring the motor. In addition, the polarity of the speed measurement can be inverted.
- Sensor latency compensation and advanced noise filtering is applied to all sensor types. Please contact Sevcon for recommended settings for these items.
- If using a PMAC motor, the absolute offset of the encoder is configured in 4630_h, sub 4. A range of +/-180° may be input. Units are in electrical degrees.
- The software has provision for an approximate auto-detection of the encoder offset using a low-speed rotation test, provided the motor is unloaded. Please contact Sevcon for information on this feature.

Resolver configuration

Please refer to page 5-8 for electrical information on the resolver subsystem.

All resolver specific settings are located in 4632_h.

- The controller can support multi pole-pair resolvers - . Please see page 5-8 for information on pole pair values other than 1.
- The controller supports a range of resolver gains by manipulating the magnitude of the excitation waveform. The phase of the resolver must also be compensated for, in order to achieve optimum performance. Resolver gain and phase settings can be auto-detected by the HVLP. Please contact Sevcon for information on tuning resolver magnitude and gain settings.

Sin/Cos configuration

Please refer to page 5-7 for electrical information on the sin/cos subsystem

Sin/cos specific settings are located in 4633_h.

- The controller can support sin/cos encoders with multiple waves-per-revolution. Please see page 5-7 for information on waves-per-revolution values other than 1.
- The minimum and maximum voltage levels of the sin/cos encoder must be entered in 4633.
- To assist in configuration, the controller provides real-time feedback of the minimum and maximum sin/cos voltage levels. A rotating motor is required to determine these thresholds.
- To account for any sin/cos sensor latency, please ensure the correct latency is set in the general encoder settings (4631_h). Please contact Sevcon for recommended latency settings for specific sin/cos sensors.

UVW configuration

Please refer to page 5-6 for electrical information on the UVW subsystem. Please note that UVW is a differential-input 5V system requiring a 5V differential signal pair for each UVW input.

For a PMAC motor, a full 360º sequence of UVW pulses must be present for each electrical cycle of the motor.

UVW specific items are located in 4634_h.

- If the UVW inputs are inverted (active-low) then this inversion can be accounted for in the UVW settings. Alternatively the encoder sequence can be inverted in the general encoder settings (4631h).
- If significant latency is present, this can be accounted for in the general encoder settings (4631h). Please contact Sevcon for recommended latency settings.

AB configuration

Please refer to page 5-5 for electrical information on the AB subsystem. Please note that AB is a differential-input 5V system requiring a 5V differential signal pair for each UVW input.

Please note, AB is not suitable for PMAC motors, where absolute position is required.

AB specific items are located in 4635_h.

- The pulses per revolution must be set to match the encoder..
- If the AB inputs are inverted (active-low) then this inversion can be accounted for in the AB settings. Alternatively the encoder sequence can be inverted in the general encoder settings (4631h).

Digital inputs

The state of the digital inputs can be monitored using I-Protocol.

Digital inputs are interpreted as both active high and active low. The value returned via the I-Protocol is individually configurable.

Digital input status' are constantly monitored and interpreted and are output using the I-Protocol as "active", "inactive" or "error".

To configure digital inputs:

- Set active high/low logic at 4680_h.
- Set digital input polarity at 6802_h. This is used to configure normally closed/open switches.

The digital inputs thresholds vary according to the KL15 supply voltage:

4680h set Active high

Input considered active when rises above	0.775 * V_{KL15}
Input considered inactive when falls below	0.759 * V_{KL15}

4680h set Active Low

Input considered active when falls below	0.242 * V_{KL15}
Input considered inactive when rises above	0.259 * V_{KL15}

Analog inputs

The analogue input voltage are read from I-Protocol messages, values are 12bit fixed point values with a resolution of 1/256 V/bit (4:8 bit format), with 0xfff used to indicate that a channel is not used.

There are 2 variable analog supplies at pins H1 and F2.

Analogue outputs may also be controlled using the I-Protocol. Refer to the I-Protocol documentation for more information. However the encoder supply is reserved by the encoder manager, and thus should be set through the encoder configuration parameters and NOT set via I-protocol.

The following table summarises the analog inputs and any special features:

Name	Pin	Usage
Analog Input 1a	K1	Input from external voltage source or 3-wire pot wiper. Use pin H1 as supply for 3-wire pot.
Analog Input 1b	K2	
Analog Input 2a	J1	
Analog Input 2b	J2	
Analogue Input 3 (Motor thermistor)	H2	Use for motor thermistor input or 2-wire pot input. Has internal supply.

Table 21 - Analog inputs

Wire-off detection

Enable wire-off detection at 46C0h to 46C2h. For each input specify the allowable range of input voltages. To disable, set the ranges to maximum.

Motor thermistor input

You can connect a thermistor sensor to the Motor thermistor input.

Type	Specification
PTC Silistor	Philips KTY83, PT1000 etc.

Table 22 - Motor thermistor input

To setup go to object 4620_h:

- Configure as none, thermistor or CAN based thermistor
- Program the resistance / temperature profile in $461F_h$. This is an 8 point table which allows a programmable thermistor table to be configured.
- For CAN based thermistors, the temperature can be loaded into object 4600_h sub 16 using an RPDO.

Read the measured motor temperature (PTC) or switch operation at object 4600_h.

Digital (contactor) outputs

There are 2 digital outputs which can be used to control loads such as contactors, horns, lights, etc.

Error control



It is important that analogue outputs on nodes other than the master must have appropriate error configuration to protect against CANbus faults. This section explains how to configure the outputs to go to a safe state in the event of a CANbus fault. It is the installer's responsibility to define what a safe state is for each output.

In a CANopen network, the slave node on which the analogue (contactor) outputs is different to the VCU node which calculates the output value. If the CANbus fails, the VCU is no longer able to control the slave outputs. In this situation, the outputs may need to change to a safe value. This is achieved with error control.

To configure error control:

- Set each output at object $6C43_h$ to use its last set value or the value at $6C44_h$ if the CANbus fails.
- Set values if needed at $6C44_h$ for each output. These values are 32-bit integers, but the bottom 16-bits are ignored. The top 16-bits give the error value in 1/256 V/bit.

Some examples of typical configurations may be:

- Electro-mechanical brake on slave node. If CANbus communication is lost, it may be desirable to apply the electro-mechanical brake on the slave device. In this case, enable error control in $6C43_h$ and set the error value in $6C44_h$ to 0.

- Power steer contactor on slave node. If CANbus communication is lost, it may be desirable to leave the power steer output in its previous state. In this case, disable error control in $6C43_h$.
- CANbus communication error lamp on slave node. If CANbus communication is lost, it may be desirable to activate an output on the slave device. In this case, enable error control in $6C43_h$ and set the error value in $6C44_h$ to an appropriate voltage for the lamp.



The above examples are for illustration purposes only. It is the responsibility of the installer to decide on the required state for each output in the event of a CANbus failure.

Vehicle performance configuration

If your vehicle system does not include a VCU capable of providing raw motor control commands via H-Protocol, then the vehicle application sub-systems within the controller can be utilised to provide these signals for you.

Vehicle sub-systems such as TracApp and PumpApp can take in external signals using I-Protocol, and generate the H-Protocol commands appropriate for driving the vehicle.

TracApp is capable of providing drive signals for a range of vehicle types, covering on-road and industrial applications.

Requirements

In order to use TracApp, the following signals must be provided as a minimum:

- An analogue throttle signal
- Two digital signals, forward and reverse, used to select the drive direction

Depending on the application, other signals will be required in order to maintain safe operation. For instance, an FS1 switch may be required by the application in order to validate the throttle position.

Vehicle application subsystems can be enabled by setting Object Dictionary index 5800h to enable master functionality. Note, on a multi node system, it is only necessary to enable this on one node, as this will be providing H-Protocol drive commands for all other nodes via CAN.

For information regarding how to map physical H-Protocol and I-Protocol inputs and outputs to vehicle functions, refer to Sevcon Application Note “Mapping J1939 Signals to Vehicle Application Subsystems”.

Safety Interlocks

FS1

The FS1 switch is normally part of the throttle assembly. It closes when the throttle is pressed. The throttle voltage is ignored until FS1 is closed.

FS1 features are configured at 2914_h:

- SRO (static return to off): inhibits drive if FS1 is closed for the SRO delay without any direction (forward or reverse) being selected.
- FS1 recycle: forces the operator to lift their foot off the throttle before allowing drive after a direction change.

Deadman

The deadman switch operates similar to the FS1 switch, whereby, it inhibits drive until it is active. However, the deadman switch applies the electro-mechanical brake immediately on deactivation, whereas FS1 waits for the vehicle to stop before applying the brake.

Seat

The seat switch indicates operator presence on the vehicle. Drive is not allowed if this switch is open. If the seat switch opens during drive for a period longer than the seat switch delay, a fault is set, disabling drive. To clear a seat fault, close the seat switch, open FS1 and deselect the forward/reverse switch.

Set the seat switch delay at object 2902_h.

Handbrake

If mapped to a digital input, the handbrake switch inhibits drive if the vehicle handbrake is applied. Controlled roll-off detection is still active when the handbrake is applied in case the brake fails.

Sequence Fault Masking

If an application does not require it, sequence fault checking can be disabled on selected drive inputs. This is set at 2918_h.

Similarly, drive inputs can be masked when clearing drive inhibit faults. This is set at $291A_h$.

⚠ CAUTION: These masks must only be applied if the application has other adequate means of protection. It is the responsibility of the installer to ensure this.

Throttle

General

Sevcon's TracApp 'VCU' type functionality is compatible with analogue input signals that are provided via I-Protocol. To setup throttle inputs see 'Analogue' on page 6-13.

⚠ Inputs with wire-off detection should be used for the throttle input to detect wiring faults. This is especially important if a wire-off sets maximum throttle. See section Analog inputs (page 6-13) for more information. Configure throttle wire-off detection threshold at 2910_h, 21.

Setup the characteristics of the throttle at 2910_h, sub-indices 2 to 21.

Define the throttle voltage input: this is the relationship between the throttle voltage and the throttle value. Separate relationships can be specified for forward and reverse. Each

relationship has two points, a start and an end. The points are configured differently for standard and directional throttles as shown in Figure 31 and Figure 32 respectively.

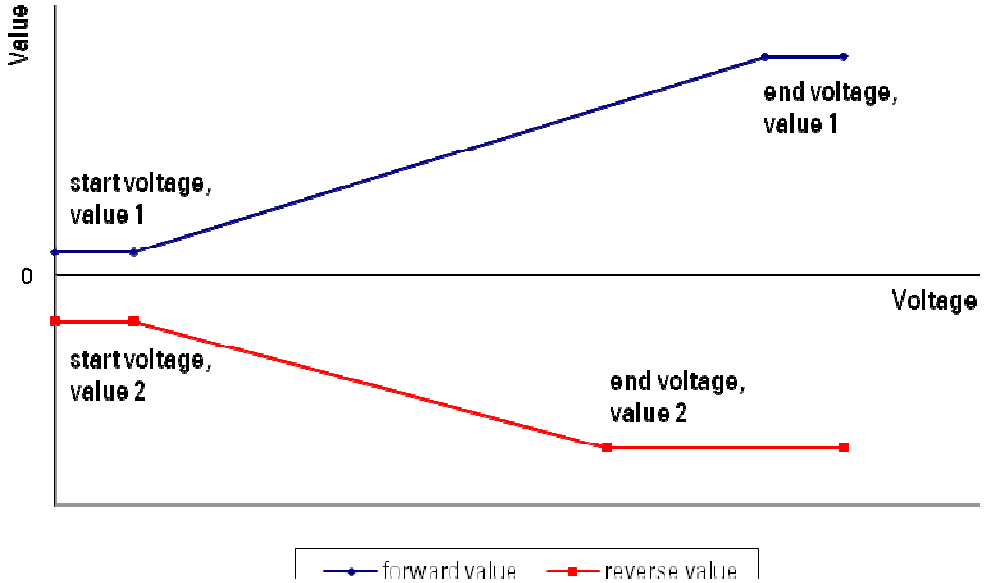


Figure 31 - Standard throttle configuration

Note: If the reverse characteristic is the same as the forward characteristic, just set all the reverse throttle parameters to 0 in 2910_h.

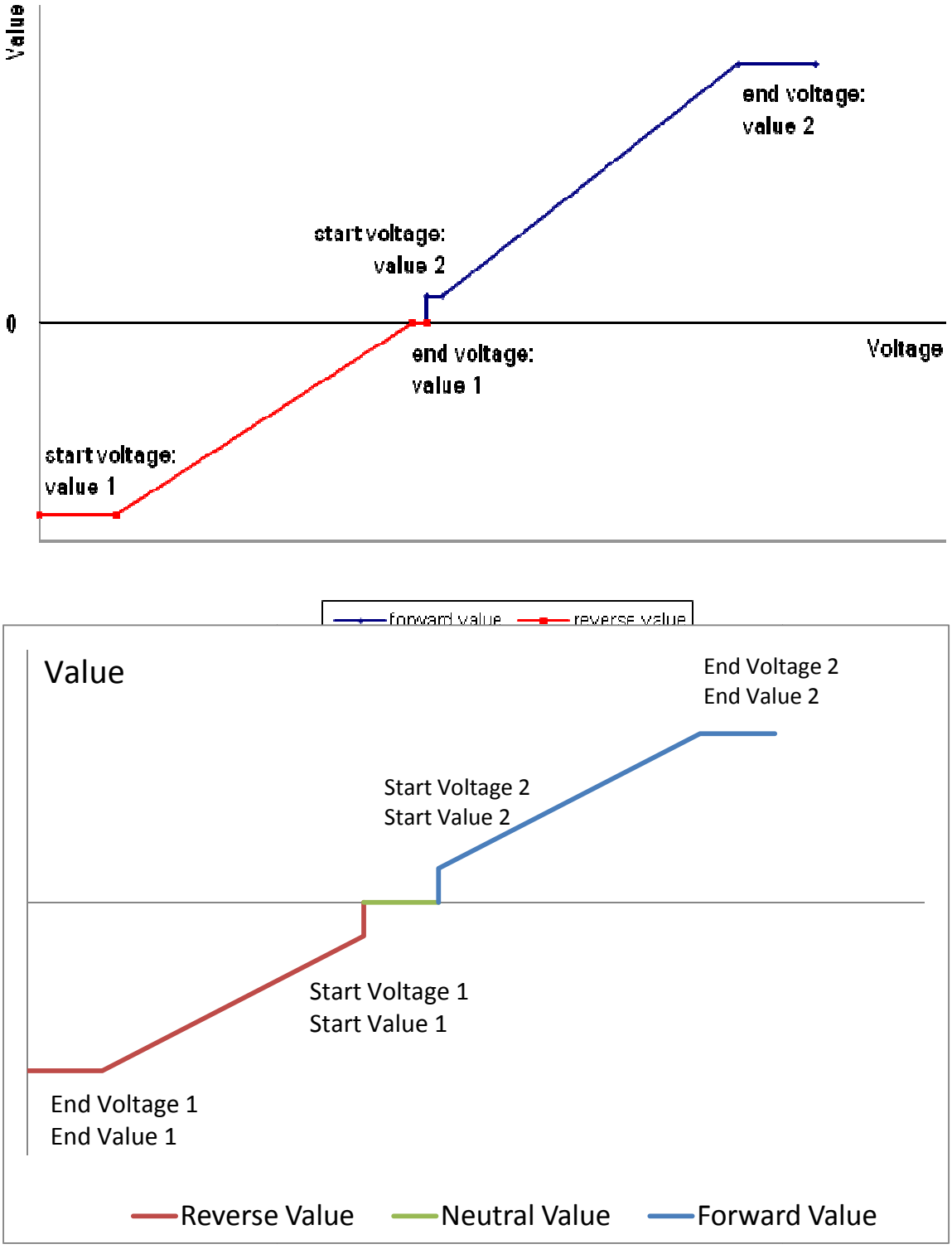


Figure 32 - Directional throttle configuration

Define the input characteristic: this is a profile to the throttle value and can be linear, curved, crawl or user-defined as shown in Figure 33. The curved and crawl characteristics give greater throttle control at low speeds.

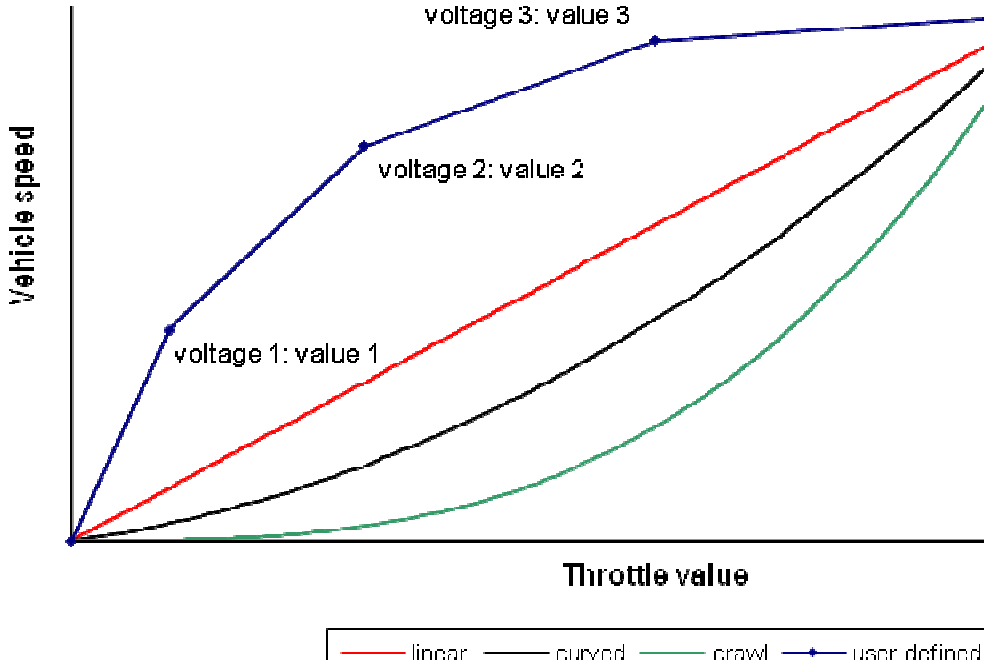


Figure 33 - Input characteristics

The throttle value calculated from the voltage can be read at 2620_h.

Dual Throttle Inputs

Single and dual throttle inputs are supported.

Single throttle inputs are normally used with other interlock inputs (e.g. FS1, deadman, etc) and use a single input voltage to determine driver demand.

Dual throttle inputs use two separate input voltages, each of which is converted to a throttle value using 2910_h, sub-indices 3 to 6 (throttle input 1) and sub-indices 7 to 10 (throttle input 2). If the throttle values differ by more than 5%, a throttle fault is set and the system will not drive.

To enable dual throttle functionality, map a second analogue input to 2224_h. The throttle value for the second throttle input can be read at 2626_h.

Dual throttle systems allow a virtual FS1 feature, which can be used instead of an actual FS1 switch. This feature can be enabled on dual throttle systems using 2910_h, 1.

CAUTION: The voltage input characteristics of the two analogue throttle inputs must be different.

Creep Torque

Set creep torque at $291f_h$. Creep torque allows a small amount of torque to be applied as soon as the throttle is closed. This can be used on some vehicles to overcome the friction required to achieve initial vehicle movement.

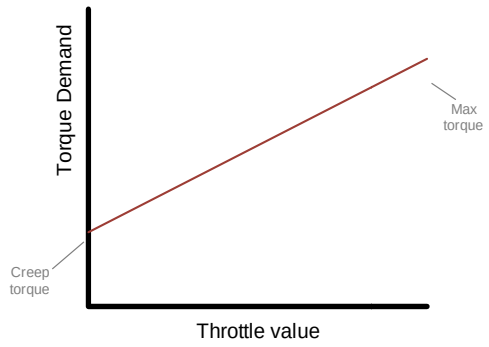


Figure 34 - Illustration showing behaviour of creep torque

Increasing the creep torque level can improve how the vehicle feels when drive is first selected and the vehicle starts to move. However, too much creep torque can make the vehicle uncontrollable at low speeds.

CAUTION: Creep torque will be applied as soon as drive is selected and the throttle is closed. Do not increase the creep torque value to a level that would cause unexpected high levels of torque output for comparatively low levels of throttle push. If in doubt, set the creep torque level to 0%.

Driveability Features

CAUTION: These features are used to configure how the system uses throttle information and how it handles speed limits (in torque mode). The installer must ensure these features are configured appropriately.

Set the following driveability features at $2910_h,1$:

- Enable/disable proportional braking. If enabled, the braking torque during direction braking is proportional to the throttle.
- Enable/disable directional throttle. If configured as a directional throttle, the throttle voltage indicates the direction as well as the speed demand. This removes the need for forward and reverse direction switches.
- Proportional speed limit enable/disable. If enabled, speed limit is proportional to the throttle, otherwise speed limit is fixed at the forward or reverse maximum speed. Only used in torque mode.

WARNING: Proportional Speed Limit is not recommended for on-highway vehicles as it can cause the traction motor/wheel to remain locked or brake severely if the wheel is momentarily locked due to loss of traction on a slippery surface and/or mechanical braking.

- Braking directional throttle enable/disable. If enabled, a directional throttle can be used to demand a drive or braking torque in conjunction with the direction switches. Only used in torque mode.
- Handbrake fault. If enabled, a handbrake fault is set when a direction is selected whilst the handbrake input is active.
- Proportional speed limit during braking enable/disable. If enabled, speed limit is proportional to throttle only in drive states. Maximum speed limit is allowed in braking states. Only used in torque mode and for open loop slip motors. Proportional speed limit above must be enabled for this option to have any effect.
- Driveability Consolidation. Normally, driveability profiles are only used to reduce vehicle performance, however, if this is enabled, an active driveability profiles over-writes the baseline. This allows vehicle performance to increase when a profile is active. Note, that this feature is not available in all software builds.
- Allow step change in steering angle. If enabled, steering angle can change instantly with steering voltage. If disabled, steering angle is rate limited to 90°/s, which prevents sudden steering angle changes in the event of a steering sensor wire-off.
- Virtual FS1 enable/disable. If enabled, this sets up a virtual FS1 feature on systems with dual throttle inputs configured.
- Positive (absolute) Steer Angle only enable/disable. If enabled steer angle is always positive in both directions. Used for single traction systems with steering cutback. Motor mapped must be left motor
- Separate Seat Regen Braking enable/disable. If enabled, separate regen setting is activated for regenerative braking with operator off the seat. Seat Regen settings are set at 2928_h.
- Slave Left Motor Speed Inversion enable/disable. Inverts left motor slave actual speed when slave inversion switch activates.
- Proportional speed limit during drive enable/disable. If enabled, speed limit is proportional to throttle only in braking states. Maximum speed limit is allowed in drive states. Only used in torque mode and for flux vector motors. Proportional speed limit above must be enabled for this option to have any effect.

Set the following driveability features at 2910_h,22:

- Direction selected indication in controlword is latched enable/disable. If enabled the selected direction bit in the control word holds the previous selected direction and doesn't return to neutral. This will help prevent the vehicle from rolling back uncontrolled on an incline if the throttle is pressed very briefly when hill-hold or controlled roll-off is active.
- The latching on the seat regen braking enabled/disable. If enabled (default) the seat regen braking state can only be exited by closing the seat switch, then changing the direction selection, for example from Forward to Neutral. If disabled, the seat regen braking state will be exited when the seat switch is closed.
- Brake feathering during seat regen braking enable/disable. If enabled (default) brake feathering will be applied to the braking torque during seat regen braking state. If disabled it will use the level set in 2928_h to 0 RPM.

An s-curve profile can be applied to the speed target (in speed mode) or maximum speed (in torque mode). This can be set at $290A_h$.

Acceleration and braking

See 'Driveability profiles' for more information on page 6-24.

Some vehicles can exhibit shock due to the rapid reversal of torque after a direction change. 2909_h can be set to force the vehicle to remain stationary for a period before driving in the new direction.

To prevent early exit from neutral braking, a debounce timer can be set at $290D_h$. Neutral braking only finishes when the vehicle has been stopped for longer than this time. This can help prevent early exit of neutral braking due to motor oscillation caused by under damped suspension.

On vehicles with gearbox meshing issues, a slower rate of torque ramp up at low speeds can be configured at $291C_h$. This slow rate of change of torque lessens shock due to gear meshing. Used in torque mode only.

Brake feathering reduces neutral and foot braking torques as the vehicle speed approaches 0 to prevent any roll-back in the opposite direction. This is set at $290E_h$. Use $290E_h$,1+2 to set start and end speeds for feathering, $290E_h$, 3 to configure the final torque at the end of braking and $290E_h$, 4+5 to configure entry and exit ramp rates for feathering torque to prevent disturbance in torque when changing between drive and braking conditions. Used in torque mode only.

Footbrake

The controller can use a switch or analogue voltage as the footbrake input. If a footbrake switch is mapped, it applies maximum foot braking when the switch is closed. The footbrake switch object (2130_h) must be mapped to a digital input.

If the footbrake input is an analogue voltage, configure the voltage levels in the same way as the throttle. The footbrake voltage (2221_h) must be mapped to an analogue input.

Configure the characteristics of the footbrake at 2911_h:

- Drive/foot braking priority. If the throttle and footbrake are pressed at the same time, this setting determines whether the system attempts to drive or brake.
- Minimum speed for braking. Foot braking stops when the vehicle speed drops below this level.
- Footbrake voltage input and Input characteristic. These settings are similar to those for the throttle. Refer to the Throttle section above for more information.
- Footbrake priority timeout allows the configured priority to change after a timeout period. Setting zero will disable this feature.

The footbrake value calculated from the voltage can be read at 2621_h.

Steering inputs – twin driving motor systems



CAUTION: Loss of steering information can make a vehicle operate erratically. The analogue input use for the steering sensor should have suitable wire-off protection configured.

Twin motor systems, which use the drive motors for turning, require some means of determining the angle of the steering wheel.

To do this use one of these options:

- A steering potentiometer to give an analogue voltage which is a linear function of the steering angle. The steer potentiometer voltage (2223_h) must be mapped to an analogue input.
- Four digital inputs representing 'inner left', 'inner right', 'outer left' and 'outer right'. The inner switches indicate the steering angle where torque to the inner wheel motor is removed. The outer switches indicate the steering angle where inner wheel motor changes direction. The outer switches are optional. The steer switches ($212B_h$ to $212E_h$) must be mapped to digital inputs.
- Steering angle from 3rd party CAN device. This can be received via RPDO on object 2624_h in 0.01°/bit resolution.
- To configure steering inputs go to index 2913_h in the Object Dictionary:
- Setup the voltages corresponding to fully left, fully right and straight ahead. Using this information, HVLP calculates the steering angle based on the voltage from a steering potentiometer.
- Setup the steering map. This map defines the relationship between the inner and outer wheel speeds and the steering angle. Each map has 4 user definable points as shown in Figure 35.

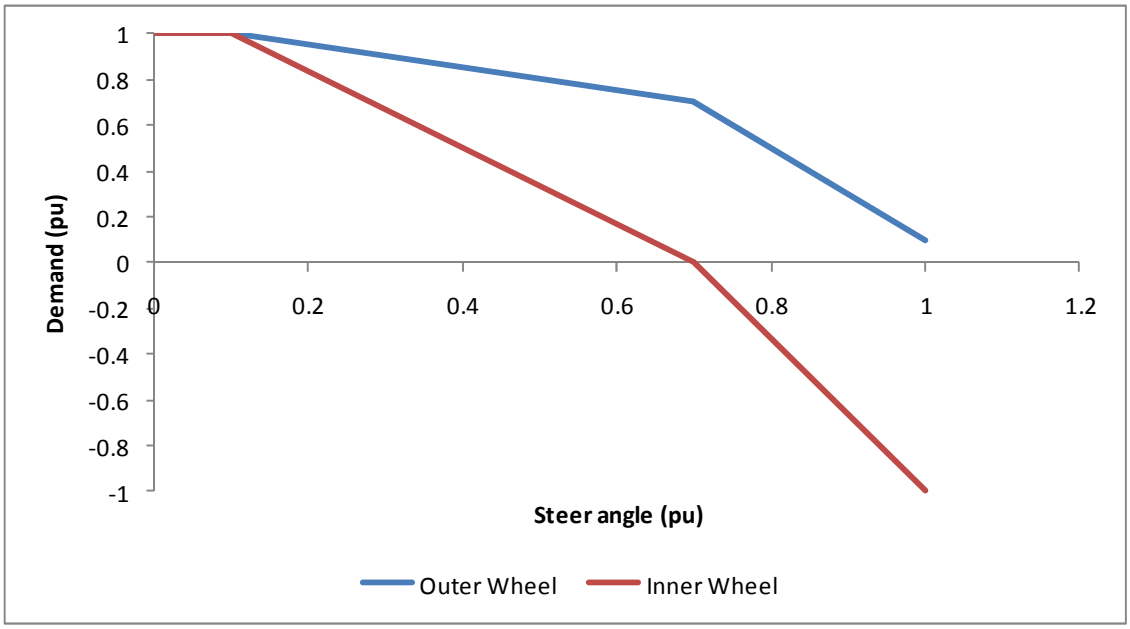


Figure 35 Graph of speed vs. steering angle

The speed and steering angle are normalized. Speed is normalized to maximum vehicle speed and the steering angle to 90°.

In speed mode, outer wheel speed target and maximum torque is scaled according the outer wheel map. Inner wheel speed target and maximum torque is scaled to the outer wheel demands according to the inner wheel map.

In torque mode, both inner and outer wheel maximum speeds are scaled according the outer wheel map. The outer wheel target torque comes from the throttle. The inner wheel target torque is scaled to the outer wheel actual torque according the inner wheel map.

In object 2913_h, 0 to 1 is represented by values in the range 0 to 32767. The inner wheel is scaled according to the outer wheel. Where a demand (pu) of -1 is shown at 90° for the inner wheel, this means the inner wheel demand will be equal and opposite to the outer wheel.

The calculated steering angle can be read at 2623_h. An angle value of -32767 indicates full steering to the left, +32767 full steering to the right and 0 is straight ahead.

If steering switches are used instead of a steering potentiometer, only part of the steering map is used as shown in Table 23.

Value	Description
2913_h,9	Outer wheel speed during inner wheel cutback
2913_h,11	Outer wheel speed during inner wheel reversal
2913_h,17	Inner wheel cutback speed
2913_h,19	Inner wheel reverse speed

Table 23 Objects to set when using steering switches

NOTE: During a turn the inner wheel speed is slowed by power reduction instead of braking to prevent the outer wheel motor working against the inner wheel motor.

Driveability profiles

WARNING: Ensure driveability profiles are disabled when not required. Activation of a driveability profile can cause driving parameters to change.

Driveability profiles allow you to set maximum values for speed, torque, acceleration and deceleration for use in a range of operational situations. In addition, in torque mode, there are ramp rates for speed limits as well. **Erreur ! Source du renvoi introuvable.** and Figure 37 show the change in speed and torque target under various driving conditions over a period of time.

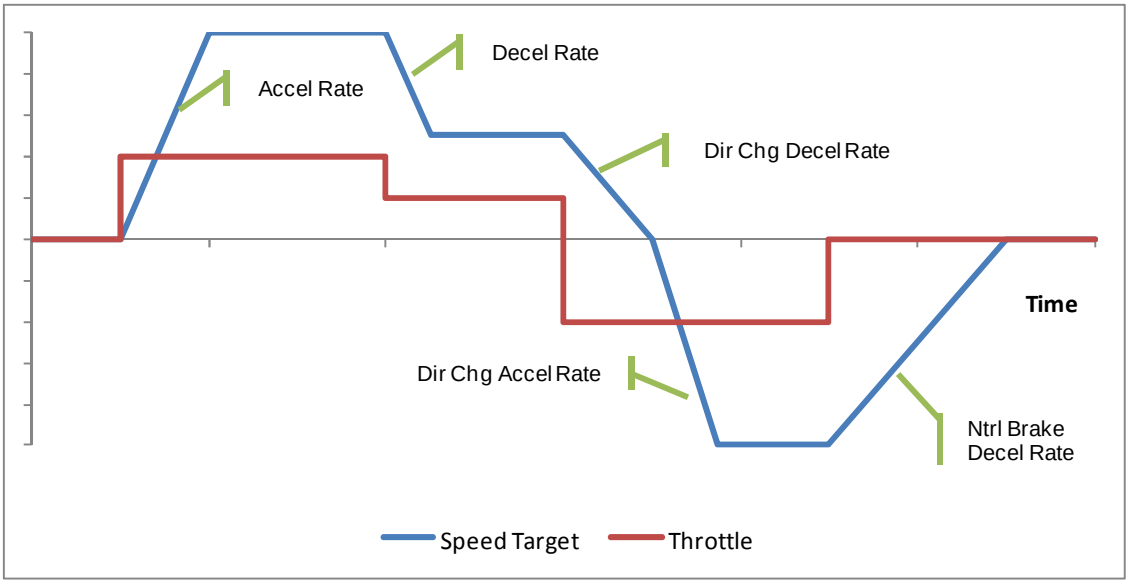


Figure 36 Speed mode acceleration/deceleration

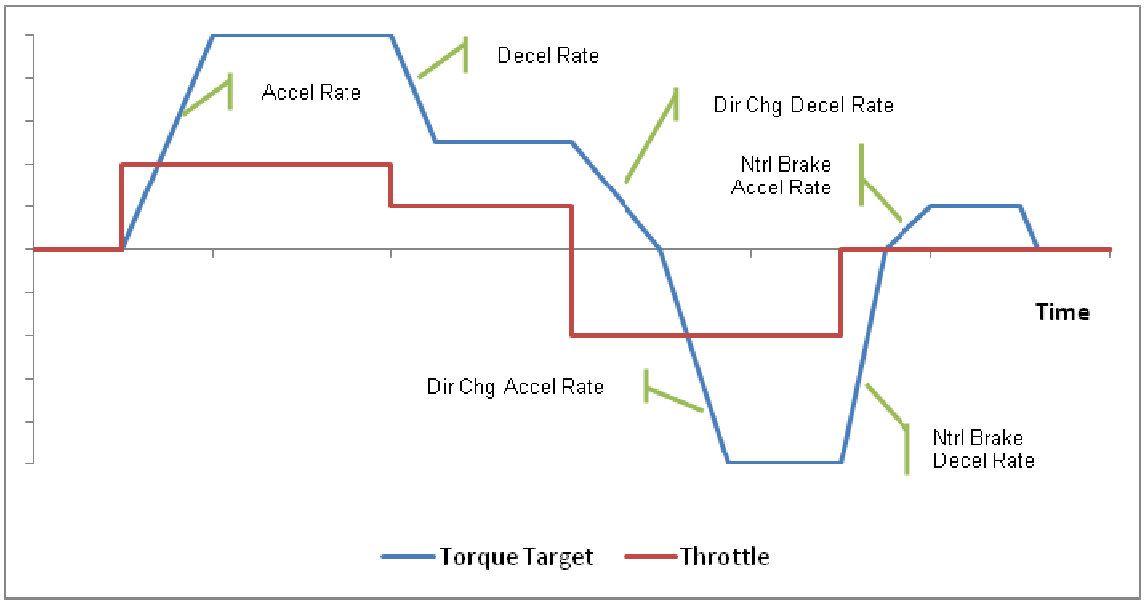


Figure 37 Torque mode acceleration/deceleration

In Torque mode, the acceleration and deceleration rates control the rate of change of torque. In Speed mode, the acceleration and deceleration rates control the rate of change of speed.

You can select reverse while driving in the forward direction with your foot still on the throttle. In this situation the controller applies braking in the form of a direction change deceleration rate down to zero speed. It then applies a direction change acceleration rate to increase the vehicle's speed in the reverse direction up to the set maximum speed as shown above.

Configure the following drivability profiles to suit your application (each containing the same set of parameters):

- Traction baseline profile: the default and highest set of values (2920_h).
- Drivability select 1 profile: invoked when drivability select 1 switch is active (2921_h) or an alternative trigger is active (see below).
- Drivability select 2 profile: invoked when drivability select 2 switch is active (2922_h) or an alternative trigger is active (see below).

The traction baseline profile contains the default maximum values. All of the remaining profiles apply lower, modifying values to the baseline profile. BDI and service profiles, when configured, are automatically applied by the software under preset conditions. For example you may want to limit the acceleration and maximum speed of a vehicle when the battery gets low to maximize the operating time before recharge. The remaining profiles are applied by the driver with a switch.

profiles can also be invoked by internal software triggers, such as BDI low, service required or low speed. These can be selected to suit specific application requirements. Set the profile triggers in 2931_h.

NOTE: Where more than one profile is active, the lowest value(s) are used by the software.

Speeds in driveability profiles are scaled according to the vehicle gear ratio (2915_h). This is used to convert speed in RPM to any other preferred unit such as KPH or MPH. To remove this scaling and leave driveability profile speeds in RPM, set 2915_h,3 to 1.

Torques in driveability profiles are in 0.1%/bit resolution. These are converted to Nm using the motor rated torque value at object 6076_h.

Ramp rates in driveability profiles are in either RPM/s for speed mode, or %/s for torque mode. In speed mode, RPM/s becomes “User Defined Units” / s if the gear ratio is used to rescale the driveability profile speeds.

Speed limit ramp rates are only used in torque mode and are in RPM/s (or user defined units / s).

WARNING: In addition to the speed limit ramp rates in the profiles, $291E_h$ can be used to configure safety limits on speed limit ramping. The installer should set these ramp rates to suitable levels to ensure speed limits cannot ramp faster than could actually happen on a vehicle. This can protect against harsh braking if traction wheels are momentarily locked.

Smoothing Output Torque

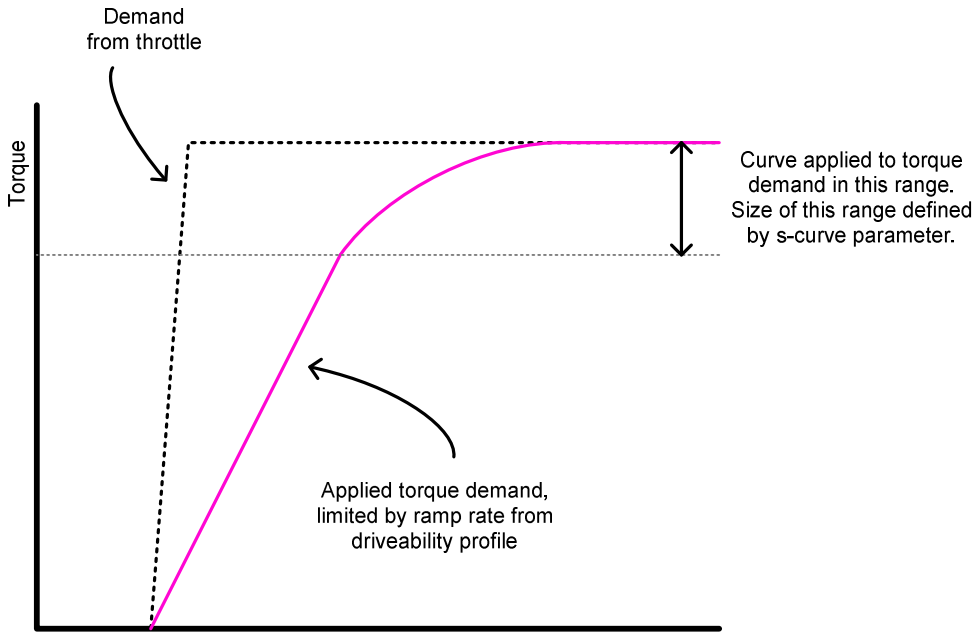


Figure 38 - Illustration of s-curve on torque demand

The controller provides an option to insert an s-curve to the torque demand as it approaches the value requested by the throttle. This curve feature can serve two purposes:

The amount of curve applied can be configured in the object dictionary at index 2933_h. Generally, applying a greater curve range will result in smoother drive, but too much can lower vehicle response. S-curve is applied according to this table:

Value in 2933_h	S-curve behavior
0	None applied
1	Curve applied within 0.2% of target
2	Curve applied within 0.4% of target
3	Curve applied within 0.8% of target
4	Curve applied within 1.6% of target
5	Curve applied within 3.2% of target

This table continues, with each increment in 2933_h doubling the range over which the curve is applied.

Scenarios

For certain vehicle types, particularly on-highway vehicles or electric motorcycles, the possibility of wheel locking during drive must be considered.

During braking, the controller will maintain a speed limit to ensure the vehicle does not over speed if entering braking whilst travelling downhill. If proportional speed limit is set then the speed limit will follow actual speed toward zero whenever actual speed is dropping rapidly, usually due to some external influence such as application of mechanical brakes.

If the brakes are applied too harshly, then there is possibility to lock the drive wheels. In these circumstances, the normal reaction of the driver is to release the brake to allow the wheels to rotate again. The correct operation of the controller in this scenario is to allow the wheels to continue to rotate, and not impose a speed limit.

The maximum rate at which the speed limit can increase or decrease is specified in object $291E_h$. By limiting the rate at which the speed limit can decrease, we can ensure that if the brakes are released after they had locked the drive wheels, the controller's speed limit will allow them to rotate again. The operation of this is shown in Figure 39 au-dessous.

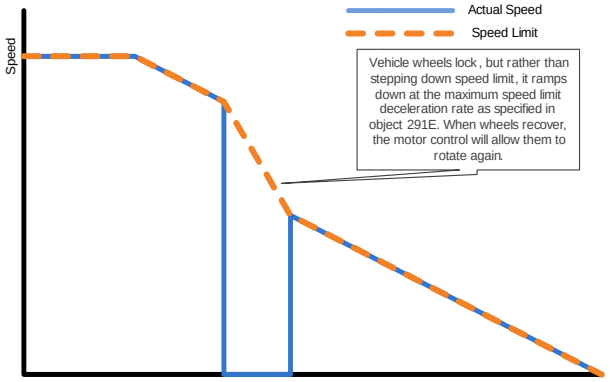


Figure 39 - Example of behaviour of speed limit when drive wheels are locked

WARNING: It is important to consider the behaviour of the vehicle under all drive conditions, including when traction is lost due to locking of the drive wheels. When testing a vehicle, check that the vehicle behaves in a safe manner when performing harsh braking on low-friction surfaces such as gravel.

Controlled roll-off

WARNING: Controlled Roll-Off is not recommended for on-highway vehicles as it can cause the traction motor/wheel to remain locked or brake severely if the wheel is momentarily locked due to loss of traction on a slippery surface and/or mechanical braking.

Controlled roll-off limits a vehicle to a slow, safe speed if it starts to move without any operator input. Primarily, it is to prevent uncontrolled movement if a vehicle's brakes fail on an incline. Controlled roll-off operates whether the operator is present or not.

Configure the following at object 2930_h:

- Enable/disable controlled roll-off
- Set a roll-off maximum speed
- Set a roll-off maximum torque

Alternatively, HVLP can apply an electromagnetic brake if one is mapped and roll-off is detected. Refer to 'Electro-mechanical brake' on page 6-33 for more information.

Hill hold

WARNING: Hill Hold is not recommended for on-highway vehicles as it can cause the traction motor/wheel to remain locked or brake severely if the wheel is momentarily locked due to loss of traction on a slippery surface and/or mechanical braking.

A vehicle on a hill can be held at a standstill for a configurable time when the operator selects neutral. At the end of this time or if the seat switch indicates the operator is not present, hill hold terminates and the vehicle can start to move if parked on an incline. If enabled, the system will enter controlled roll-off after hill hold.

You can set the hill hold delay at object 2901_h. Set the hill hold delay to 0 to disable this feature.

In speed mode, drive torque is disabled whilst neutral braking to stop. However, drive torque must be re-enabled when entering Hill Hold to allow torque to be applied to hold on the incline. Set the speed to re-enable drive torque at 2908_h.

Inching

WARNING: Ensure inch switches are only mapped to digital inputs when required. Activation of these inputs can cause a drive condition to occur.

Inching allows an operator to manoeuvre a vehicle, at low speeds, towards a load. Inching can be initiated with one switch. A time-out is used to prevent the vehicle from continuing to drive indefinitely if the switch gets stuck or goes short circuit.

To configure inching:

- Ensure forward and reverse inching switches have been mapped to two digital inputs.
- Specify an inching speed (0% to 25% of the full speed of the vehicle) at 2905_h sub-index 1. This is either a speed target in speed mode, or maximum speed in torque mode.
- Specify an inching throttle (0 to 100%) at 2905_h sub-index 3. This gives a torque target in torque mode. This is not used in speed mode.
- Specify a time-out (0.1 s to 5.0 s) at 2905_h sub-index 2.

Belly Switch

WARNING: Ensure the belly switch is only mapped to a digital input when required. Activation of this input can cause a drive condition to occur.

The belly switch is normally connected to the end of the tiller arm on class 3 vehicles. When activated it forces a drive condition in forward at a user specified throttle value and maximum speed for a specified time.

To configure belly:

- Ensure the belly switch is mapped to a digital input.
- Specify the maximum belly speed at $290C_h$ sub-index 2.
- Specify a belly throttle at $290C_h$ sub-index 1. This will determine the torque demand in torque mode or speed demand in speed mode.
- Specify a belly time out at $290C_h$ sub-index 3. The belly function will cease after this time has expired.

Drivability select switches

WARNING: Ensure the driveability switches are only mapped to digital inputs when required. Activation of these inputs can cause driving parameters to change.

There are two drivability select switches (2126_h and 2127_h).

To enable either of these they must be mapped to digital inputs. When they are active, the corresponding drivability profiles (2921_h and 2922_h) are applied.

See Driveability profiles (page 6-24) for more information.

Economy

The economy input is an analogue input which can be used to increase vehicle efficiency and extend battery life. It is normally controlled using a potentiometer mounted on the vehicle's dashboard. The economy voltage (2222_h) must be mapped to an analogue input.

Efficiency is improved by reducing the acceleration rate or the maximum torque.

Configure the economy input at object 2912_h as follows:

Economy function: select acceleration or torque.

Economy voltage input: These settings are similar to those for the throttle (see page 6-16).

The economy value calculated from the voltage can be read at 2622_h.

Seat Regen Braking

The seat switch can be used to override the current driveability settings with a higher value as set in 2928h. This allows the vehicle to use stronger braking to bring the vehicle to a halt when the driver is not present. This overrides the following settings:

- Maximum torque, Acceleration and Deceleration rates in Neutral Braking
- Maximum torque, Acceleration and Deceleration rates in Footbraking

This can be configured to only be activated above a certain speed to prevent activation when the vehicle is stationary. A "Seat Warning" fault is set in this state.

Pump configuration

The controller can use a mixture of switch and analogue voltages as the pump input. In addition, the power steer function can be used as an extra input to the pump if the pump motor is required to supply pump and power steering.

NOTE: Pump motors always run in speed mode. Ensure the motor slave is also configured for speed mode in 6060_h.

General Setup

Configure the pump features at $2A00_h$:

- Inhibit pump when BDI drops below cut-out level. If already operating when the cut-out occurs, the pump will continue to operate until all pump inputs are inactive.
- Drive Enable switch and/or Seat switch input disables pump.
- Ignore Line Contactor state. Allows the pump to operate if it is not connected to the battery through the line contactor. Should be set if the pump also supplies power steering and the power steer is required to operate when the line contactor is open.
- Use Power Steer target velocity as pump input, if pump also supplies power steering.
- Enable minimum pump speed. Enable this to force the pump to run at minimum speed ($2A01_h$, 2) even when there is no trigger. Can be used to maintain minimum pump pressure.
- Pump to stop on Low Battery. Enable to force pump to stop immediately on low battery condition.
- Use power steer demand to minimum pump speed. Enable this to force the pump to use power steer demand as a minimum speed. Can be used to maintain minimum pump pressure for power steering.
- Set the pump minimum and maximum speed, maximum torque, acceleration and deceleration at $2A01_h$. The pump speed is calculated as the value from the inputs multiplied by the maximum speed.

Priority/additive inputs

Each pump input can be configured as a priority input or an additive input. When calculating the pump demand, the controller selects the demand from the highest priority active input, and then adds the demand from all the active additive inputs.

Configure priority/additive levels in $2A10_h$ and $2A11_h$, and $2A20_h$ to $2A26_h$.

Pump throttles

There are 2 pump throttle inputs, which can be configured independently at $2A10_h$ and $2A11_h$. The pump throttles allow proportional control of the pump speed.

Configure inputs as priority or additive and set the voltage levels in the same way as the traction throttle. The pump throttles must be mapped to analogue inputs.

Pump switches

There are 7 pump switch inputs. Configure each input as priority or additive and assign it a value at $2A20_h$ to $2A26_h$. The pump switches must be mapped to digital inputs.

Pump Driveability Profiles

Pumps have configurable driveability profiles. Profiles are triggered by pump driveability select switches (2152_h and 2153_h). One or more of these switches must be mapped to enable pump profiles.

Each profile allows the installer to reduce acceleration and deceleration rates, throttle and switch values and maximum torque.

Set pump driveability profiles at $2A30_h$ and $2A31_h$

Power steer configuration

General

Power steering can be provided using:

- Contactor. Map the power steer contactor drive object to an analogue output.
- Pump motor controller. Configure pump to provide power steering. Power steer demand is added to pump demand.
- Dedicated motor controller. Map power steer application motor object to motor control slave.



Power steer motors always run in speed mode. Ensure the motor slave is also configured for speed mode in 6060_h.

The power steer can be triggered by a number of events:

- Vehicle moving
- FS1 switch activating
- Direction selected.
- Seat switch activating
- Footbrake activating



CAUTION: The power steering function will always attempt to execute, even if the line contactor is open due to a fault condition. This is to ensure power steering continues to operate at all times.

Set the power steer motor speed, acceleration and deceleration at $2B01_h$. This is not required if the power steer motor is operated by a contactor.

Variable Assist Power Steering

HVLP supports a variable assist power steering algorithm which can be used to reduce the power steering speed as vehicle traction speed increases to a user configurable level. Set the reduction factor and traction speed in $2B02_h$. This allows power steering effort to be reduced as vehicle speed increases to prevent steering becoming too light.

Vehicle features and functions

Ensure you have completed the CANopen network setup, Motor Characterization, I/O Configuration and Vehicle Performance Configuration processes described above.

Contactors

Ensure voltage control has been selected (see 'Analogue' on page 6-14).

To configure any contactor:

- Set pull-in voltage, pull-in time and hold-in voltage at $2D00_h$
- Enable each output to operate at the pull-in voltage or at the maximum voltage at $2D01_h$
- If required enable each output to reduce to the hold voltage level at $2D02_h$

Line contactor

The line contactor must be mapped to an analogue output.

The line contactor is used to isolate controllers and motors from the battery during power down or in case of a serious fault. It is normally closed all the time the vehicle is powered, but it can be configured to open when the vehicle has been stationary for a period of time.

Configure line contactor dropout at object 2820_h. See also 'Contactors' above.

The controller has a capacitor pre-charge feature used to protect line contactor tips from damage due to in-rush currents when the contactor closes.

NOTE: Writing to 5180_h starts a pre-charge cycle. The pre-charge circuit can only supply enough current to charge the capacitors of one controller. Where more than one controller is present, the pre-charge circuit on each must be used. If an HVLP is configured as the vehicle master, it controls the pre-charge of all slave nodes automatically.

The pre-charge level is configurable between 50% and 90%. The level can be adjusted by object 5820_h.

The minimum capacitor voltage allowed for the line contactor to close is set in 5821_h.

Electro-mechanical brake

WARNING: Electro-mechanical brakes are not recommended for on-highway vehicles as they can cause the traction motor/wheel to remain locked or brake severely if the wheel is momentarily locked due to loss of traction on a slippery surface and/or mechanical braking. Also, electro-mechanical brakes normally fail to the applied state, meaning any loss of power, or wiring fault can cause the brakes to be applied.

The electro-mechanical brake must be mapped to an analogue output.

Set the conditions under which it is applied at 2903_h.

The brake can be applied when the vehicle stops or when roll-off is detected. If the brake is configured to apply when the vehicle stops, it is not applied until the vehicle has been stationary for more than the brake delay time. The brake also can be configured to not apply if the vehicle speed exceeds a minimum, configured in 2903_h,6, during the brake apply timer period. This can be used to prevent the electro brake being applied whilst the vehicle is moving.

In the event of a severe fault the brake can be configured (via 2905_h,5) to apply immediately or to wait until the vehicle to stops before applying the brake. In the event of an encoder fault however brake remains released.

WARNING: In the event of an Encoder Fault on a *slave node* causing a perceived zero speed by the master traction controller, the brake will apply regardless of the setting in (via 2905_h,5). This caveat only applies to multi node systems.

It is possible to detect an electrobrake wire off by configuring a minimum brake coil current in 2903_h,4. Should the coil current drop below this value whilst energised, then the controller will set a Drive Inhibit Fault and neutral brakes to a stop. The fault clears when neutral is selected, if the brake is commanded to be applied, or neutral is selected and the current exceeds the minimum value.

WARNING: It is not possible to configure electro brake wire off on slave nodes, where the brake is released via a CANopen PDO.

To prevent vehicle roll away on inclines, the electro-mechanical brake normally does not release until the traction motor(s) are producing torque. This feature can be disabled using 2903_h,3.

External LED

This mirrors the operation of the controller's on board diagnostic LED. The external LED can be mapped to an analogue output to drive a lamp on a vehicle dashboard.

Alarm buzzer

The alarm buzzer can be mapped to an analogue output.

Configure the alarm buzzer output, if required, to be activated by one or more of these conditions at 2840_h:

- forward motion or forward direction selected
- reverse motion or reverse direction selected
- faults other than information faults
- controlled roll-off
- BDI low.

A different cadence for each of the above conditions can be configured.

Brake Lights

A brake light output object is available and can be mapped to an analogue output. The brake lights will illuminate whenever the footbrake is pressed (providing either an analogue or digital footbrake input is available) or the system is in direction change braking.

Horn

Ensure a digital input switch is mapped to the horn switch, and an analogue output is mapped to the horn output.

Vehicle Speed Calculation

The controller can be configured to calculate vehicle speed from motor speed by setting a configurable ratio between these values in object 2915_h.Calculated vehicle speed can then be transmitted to the CANbus for use with compatible displays, or used to keep a log of total distance travelled by the vehicle.

The convention is to calculate vehicle speed as a signed number in 12.4 format in kph. This is required for compatibility with ClearView displays and the odometer. In case a custom display is used whose units are different, a user speed calculation is also provided whose units are independent from all other features.

Two sets of ratios are provided for calculation of a vehicle and user speeds, located in object 0x2915. While vehicle speed is a signed value, the option to have unsigned user speeds is also provided (an absolute value will be calculated).

The ratio between motor speed and vehicle speed will be dependent on vehicle drive train parameters such as gearbox ratio and wheel size. However, the ratio should be calculated such than when it is multiplied by motor speed (which is in rpm), the result is vehicle speed in kph 12.4 format. To do this, the following equation can be used:

- Set the divisor (object 2915_h sub index 2) to a known speed point of the motor
- Set the multiplier (object 2915_h sub index 1) to the corresponding scaled vehicle speed for the known motor speed.

For example, if it is known that 1000rpm on the motor results in a vehicle speed of 5kph, then the divisor should be set to 1000, and the multiplier should be set to 0x0050, which is 5kph in 12.4 format.

The same process should be repeated for the user speed ratio, setting the multiplier for the user speed in the desired custom format. Both the vehicle and user speeds, objects 0x2921 and 0x2922, can be mapped to TPDO for transmission to other devices via CAN if required.

Distance Calculation

Total vehicle distance and trip distance counters are available at objects 0x29A0 and 0x29A1. Counters will operate and will be accurate provided that the vehicle speed calculated at object 0x2721 is in kph in 12.4 format. Counters are available in the following formats:

- Distance travelled in km in 24.8 format
- Distance travelled in km in 0.1km/bit increments
- Distance travelled in miles in 0.1mile/bit increments

Both distance counters objects can be mapped to TPDOs for monitoring over the CANbus, or for use by a dashboard display. The trip distance counter can be reset by holding the reset switch, object 0x217D, closed for 1 second. The total vehicle distance counter cannot be reset by the user.

Service Indication

The controller can reduce vehicle performance and indicate to the operator when a vehicle service is required. The interval between services is user-configurable.

Configure the following at object 2850_h:

- Source hours counter: selects the hours counter and is used to determine when a service is required.
- Service interval: hours between vehicle services. Can be used by the reset function (see below) or for information only.
- Next service due: Servicing is required when the source-hours counter reaches this time. This can be set manually, or automatically using the reset function; see below.
- Reset function: write to the reset sub-index at 2850_h to automatically reset the service timer for the next service. The next service due time is calculated as the source hours counter time plus the service interval.

Service profile

This is a drivability profile where you can set maximum torques, speeds and acceleration rates to be applied when a vehicle needs servicing (2925_h). See 'Driveability profiles' on page 6-24.

Traction motor cooling fan

This object can be used to drive a motor cooling fan when the operator is present on the vehicle (as indicated by the seat switch). The cooling fan object (2421_h) must be mapped to an analogue output.

Controller heatsink / motor cooling fan

An external fan to cool the controller heatsink or a motor may be connected to one of the analogue outputs. The fan will be turned on by the controller when either the heatsink temperature or the motor temperature exceed a specified temperature. The fan turns off when the nominated temperature is cold. The temperatures at which the fans should turn on and off, the analogue output to use for the fan, the fan voltage and the temperature source (heatsink or motor) can be programmed using the heatsink fan object ($5A01_h$). Note that the contactor driver outputs may be damaged if connected to capacitive loads. It is quite common for fans to incorporate capacitive elements, in which case a relay should be used to isolate the fan from the contactor driver output.



NOTE: The temperature set-point to turn on the fans should be higher than the set-point to turn off the fans

The fans will not operate if another function is configured to run on the specified analogue output.

Motor over-temperature protection

The controller protects motors from over-temperature. It maintains a motor temperature estimate and can also accept a direct temperature measurement via an analogue input (for a thermistor) or a digital input (for an over-temperature switch).

The temperature estimate is calculated by monitoring current to the motor over time. The estimate is configured at 4621_h.

The estimate is always applied, since it can detect increases in motor temperature more quickly than the direct measurement. Direct measurement is normally done on the motor casing, which lags behind the internal temperature.

Additional protection is provided for thermistor wire-off conditions. If the temperature sensor input measures a completely short circuit or open circuit signal, then the output torque will begin to reduce gradually over time. A recovery rate can also be specified should the signal become valid again. This means that wire-off situations, should they occur during drive, do not result in an instant loss of torque. Both the torque ramp down and recovery rates can be configured using object 4620_h.

Motor over-speed protection

A facility to protect the motor or vehicle power train due to damage by over-speeding is available on the controller. A maximum speed can be configured at object 4624_h. Under normal operation the controller should output braking torque to prevent the over-speeding initially, if the measured speed exceeds this limit then the controller will shut down and a fault will be set.

CAUTION: The trip speed offers a final level of protection for the vehicle mechanics, and should be set to a minimum level that would not be expected to be reached under normal operation.

Battery protection

The nominal battery voltage must be set at $2C00_h$.

Over voltage

Battery over voltage usually occurs during regenerative braking.

To provide protection set values for these parameters at $2C01_h$:

- Over voltage start cutback: the value at which the braking effort is linearly reduced to limit voltage increase.
- Over voltage limit: the value at which the controller cut-outs out. A fault is set if the voltage exceeds the cut-out voltage.

Under voltage

To prevent excessive battery discharge, set values for these parameters at $2C02_h$:

- Under voltage start cutback: the value at which the current drawn from the battery is reduced to limit voltage decrease.
- Under voltage limit: the value at which the controller cut-outs out. A fault is set if the voltage drops below the cut-out voltage for longer than the protection delay
- Protection delay: the time it takes for the controller to cut-out after the under voltage limit has been reached (2C03h).

Battery Discharge Indicator (BDI)

CAUTION: When not in use ensure the BDI function is disabled by setting the Cell Count (in $2C30_h$, 6) to 0.

Monitor battery voltage using HVLP's Battery Discharge Indicator (BDI). The BDI presents the driver with a percentage remaining charge figure and has become an industry standard in recent years.

NOTE: The BDI is not a measure of the absolute battery charge remaining and therefore we recommend you regularly check the absolute value in accordance with the battery manufacturer's instructions.

To use the BDI, configure the following parameters at $2C30_h$ in the Object Dictionary:

- Cell count: this is the number of battery cells and is normally half the battery voltage, as cells are usually 2 volts each.
- Reset voltage (V): set this to the cell voltage when the batteries have just been charged. This resets the BDI back to 100%.
- Discharge voltage (V): set this to the cell voltage when the battery is discharged.
- Cut-out level (%): this is the level at which the vehicle adopts the low battery drivability profile.
- Discharge rate (s/%): this is the rate at which the BDI remaining charge value discharges. Set to 0 to use default value of 16.8s to reduce by 1%. This default should suit most lead-acid battery types, however, this can be increased/decreased for different battery technologies.

NOTE: Setting the warning and cut-out levels to 0% disables the warning and cut-out functionality

An additional table exists at index $2C31_h$ which allows an initial BDI value to be specified at start up based on the measured battery voltage. At start up, the measured battery voltage is taken, and the corresponding BDI percentage remaining value is then used as the starting point for the BDI. After this, the BDI remaining value operates in the standard way. Note, this function is only available if the corresponding bit is set in object $2C30_h$ sub 1.

Read the percentage remaining charge value from 2790_h sub-index 1 in the Object Dictionary.

Battery Current Limit

Battery current can be limited by the controller for the purposes of efficiency or to protect batteries that are sensitive to high levels of current flow. Charge and discharge currents can be limited independently.

If limiting the discharge current flow, this can extend the time taken for the vehicle to reach top speed. Note that limiting the charge current flow back to the battery can impede the performance of regenerative braking.

Object 2870_h controls how the battery current limit is calculated. Sub-index 1 of this object can be set to one of the following values to specify how the current limit behaves:

Value	Mode
0x00	Master control of battery current limit disabled. To completely disable battery current limit, object 4623_h sub index 3 must also be set to 0x0000.

0x01	Battery current limit set by object 4623_h sub index 2 and 3.
0x02	Battery current limit controlled by compatible BMS
0x03	Battery current limit set by object 4623_h sub index 2 and 3, but the drive current limit is multiplied down by factors set in sub index 4 and 5 when driveability profiles are activated
0x04	Battery current limit is calculated to maintain a power limit as set in sub index 6 and 7.

Object 2871_h provides a means to have TracApp cut back the torque demand in situations where the battery current limit is reduced. This is useful for configurations where an external BMS can cut back the current limits to low levels. Effectively, it prevents TracApp requesting torque that cannot be achieved.

Object 4623_h shows the current limits that are in effect. Sub index 3 of this object allows you to specify the cutback aggressiveness and a measurement correction factor. Setting this to zero. Battery current flow can be monitored at object $5100_{h.}$

Note that regen currents flowing back to the battery are specified as negative numbers.

Displays

HVLP is compatible with Smartview and Clearview displays.

Clearview displays use the CANopen protocol. To use, set up TPDOs to transmit the required data for the display.

Smartview displays use Sevcon's proprietary CAN protocol. To use set the CAN baudrate to 100kHz at 5900_h, enable Smartview and select hours counter and/or real speed at $2E00_h$.

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Chapter 7: Monitoring HVLP-10/20

Reading status variables

All status information is contained either in the HVLP-10/20's Object Dictionary, or transmitted using either H-Protocol or I-Protocol messages. Most service information can be accessed using SDOs. Some can be mapped to PDOs for continuous transmission to remote nodes such as displays and logging devices.

Motor Data

The following motor data is available on H-protocol:

- Motor torque, speed, battery current
- Available drive and brake torque
- Motor temperature
- Motor control and status words

The following motor data is also available on H-protocol debug messages

- Motor slip frequency, currents and voltages
- Additional motor debug information

Customer Input Data

The following customer input data is available on I-protocol:

- Digital input states
- Analogue input voltages

Device Data

The following device data is available on H-protocol:

- Device temperature
- Power frame (DC link) voltage
- Highest priority fault

Identification and version

Read identification and version information at:

- 1008_h – Controller name.
- 1009_h – Hardware version.
- $100A_h$ – Software version.
- 1018_h – Identity object. Contains CANopen vendor ID, product code, CANopen protocol revision, and controller serial number.
- 5500_h – NVM (EEPROM) format.
- 5501_h – Internal ROM checksum.

Supply monitoring

The controller measures the following supply voltages:

- KL15 and KL30 voltages; measured at KL15 and KL30 inputs and read at 5100_h.
- Capacitor voltage; measured at the B+ terminal. Read at 5100_h and also available on H-protocol.

Hours counters

The controller supports many different hours counters for various functions. Some counters run on all units and some only run on the HVLP-10/20 configured as the vehicle master.



Hours counters are preserved with a minimum resolution of 15 seconds when the system is powered down.

The hours counters are:

- KL15 key hours: increments while the keyswitch is in the ON position (5200_h).
- Controller pulsing hours: increments when the controller is powering its connected motor (4601_h).

Logging

The controller can log events in the system (along with additional event-related information) and minimum and maximum levels of important parameters. You need different levels of access to clear the contents of the logs.

Logs are normally reset individually. However, to reset all logs at once write to 4000_h.

FIFO event logs

Events are recorded by these two separate FIFOs (first in, first out logs), which operate identically:

- System: this FIFO is 20 elements deep and is used for events such as software upgrades, user logins and some hardware upgrades (4100_h to 4102_h).
- Faults: this FIFO is 40 elements deep and is used for fault logging (4110_h to 4112_h).

At object $41X0_h$:

- Reset the FIFO
- Read its length

You can access the FIFO using objects $41X1_h$ and $41X2_h$. The FIFO index is entered at $41X1_h$ and the data is read from $41X2_h$.

Event counters

The controller provides 10 event counters at 4200_h to $420A_h$. Each event counter can record information about occurrences of one event. The allocation of event counters to events is user-configurable however HVLP-10/20 will automatically count important events in unused counters. The information recorded in each event counter is:

- The time of the first occurrence
- The time of the most recent occurrence
- The number of occurrences

Operational monitoring

At objects 4300_h and 4301_h, HVLP-10/20 monitors and records the minimum and maximum values of these quantities:

- Battery voltage
- Capacitor voltage
- Motor current
- Motor speed
- Controller temperature

Two instances of the operational monitoring log are maintained. Service engineers can access and clear the first log; the second is accessible and clearable only by Sevcon engineers. The Customer copy is normally recorded and reset each time the vehicle is serviced. The Sevcon copy records data over the controller's entire working life.

CANopen abort code

The controller will sometimes respond with a CANopen General Abort Error (08000000_h) when the object dictionary is accessed. This can occur for many reasons. Object 5310_h gives the exact abort reason as shown in Table 24.

0	None	12	Invalid value	24	Cannot read from DSP
1	General	13	EEPROM write failed	25	Peek time out
2	Nothing to transmit	14	Unable to reset service time	26	Reserved for future use
3	Invalid service	15	Cannot reset log	27	Reserved for future use
4	Not in pre-operational	16	Cannot read log	28	Reserved for future use
5	Not in operational	17	Invalid store command	29	Reserved for future use
6	Cannot go to pre-operational	18	Bootloader failure	30	Reserved for future use
7	Cannot go to operational	19	DSP update failed	31	Reserved for future use
8	Access level too low	20	GIO module error failed	32	Checksum calculation failed
9	Login failed	21	Backdoor write failed	33	PDO not copied
10	Range underflow	22	Reserved for future use		
11	Range overflow	23	Cannot write to DSP		

Table 24 - Abort Reason Codes

Faults and warnings

Introduction

In the event of a fault HVLP-10/20 takes the following action:

1. Takes action appropriate to the fault severity. This can be either derating of motor power or speed or shutdown of the power frame.
2. Sends out an EMCY message on the CANbus and updates the highest priority fault ID in the H-protocol.
3. Logs the fault for later retrieval.

Fault identification

You can identify a fault as follows:

- Pick up the EMCY on the CANbus and read the fault condition using configuration software
- Interrogate the fault on the node directly using DVT or other configuration software.

Fault list

Use DVT to access the Fault list. If you don't have DVT you can use any configuration tool as follows:

1. Object 5300_h gives information about all active faults. Read sub-index 1 to get the number of active faults. Write to sub-index 2 to select one of the active faults (0 = highest priority) and read back sub-index 3 to read the fault ID at that index.
1. Object 5610_h can be used to read a text description of the fault. Write the fault ID to sub-index 1 and read back the fault description from sub-index 2.

Upgrading the controller software

It is possible to field update the firmware of the HVLP-10/20 controller, typically using Sevcon's DVT configuration tool.

Please contact Sevcon for assistance with this process.